DEVELOPMENT OF FINITE ELEMENT SOLVER FOR LIGHT SCATTERING SIMULATION

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DECLARATION BY THE CANDIDATE

I, **LEE SHIEN WEI**, hereby declare that except where due acknowledgement has been made, the work presented in this thesis is my own, and has not been submitted previously in whole or in part, to qualify for other academic award.

The content of this graduation exercise is result of the work I have been carrying out since the official commencement date of the approved thesis project.

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ABSTRAK

Projek ini menangani jurang pelaksanaan kritikal dalam elektromagnetik pengiraan dengan membangunkan penyelesai unsur terhingga sumber terbuka untuk simulasi penyerakan cahaya menggunakan elemen tepi Nédélec bertertib tinggi dalam rangka kerja FEniCSx. Objektif utama adalah untuk mewujudkan alat pengiraan yang teliti secara matematik dan tepat secara fizikal yang mendemokrasikan akses kepada analisis penyerakan elektromagnetik sambil mengekalkan ketepatan gred profesional untuk aplikasi dalam reka bentuk biosensor dan penyelidikan nanofotonik. Kepentingan penyelidikan terletak pada merapatkan jurang antara perisian komersial yang mahal dan pelaksanaan tersuai yang mencabar secara teknikal, menyediakan komuniti saintifik dengan penyelesai yang telus, boleh disahkan, dan boleh dikembangkan yang menggabungkan ketegasan teori dengan kebolehcapaian praktikal. Metodologi menggunakan elemen Nédélec pematuhan-$H(\text{curl})$ bertertib ketiga pada jaringan segi tiga tidak berstruktur untuk mendiskritkan persamaan Helmholtz vektor harmonik masa, dengan syarat sempadan Konduktor Elektrik Sempurna dikuatkuasakan melalui kekangan Dirichlet kuat dan pemotongan domain dicapai melalui syarat sempadan penyerapan peringkat pertama. Formulasi lemah dilaksanakan menggunakan Bahasa Bentuk Bersepadu dan diselesaikan melalui pemfaktoran LU langsung melalui penyelesai MUMPS. Pengesahan dilakukan terhadap penanda aras siri Mie analitikal untuk penyerakan silinder dua dimensi dalam rejim resonans ($ka \approx 3.14$), dengan kajian penumpuan sistematik dijalankan merentasi lima jaringan yang diperhalusi secara progresif. Keputusan awal menunjukkan kejayaan pelaksanaan rangka kerja matematik, dengan visualisasi medan kualitatif mengesahkan corak gelombang pegun yang betul secara fizikal, pembentukan zon bayang geometri, dan penguatkuasaan syarat sempadan yang tepat tanpa artifak palsu. Pengesahan kuantitatif mendedahkan bahawa penyelesai beroperasi dalam rejim asimptotik stabil ($kh = 0.374 < 0.5$), mengelakkan kesan pencemaran berangka. Kajian penumpuan memberikan bukti muktamad ketepatan pelaksanaan, menunjukkan bahawa ralat berangka berkurangan secara monotonik dengan penghalusan jaringan dan mencapai kadar penumpuan empirik 2.88, dalam persetujuan cemerlang dengan ramalan teori 3.0 untuk elemen Nédélec kubik. Padanan rapat antara kadar penumpuan yang diperhatikan dan teori ini mengesahkan bahawa formulasi lemah dilaksanakan dengan betul, syarat sempadan dikuatkuasakan dengan sewajarnya, dan penyelesaian telah memasuki rejim asimptotik di mana ralat pendiskretan mendominasi ralat fasa. Penyelesai dua dimensi yang disahkan ini mewujudkan asas kukuh

untuk pengembangan masa depan kepada geometri tiga dimensi dan aplikasi multi-fizik tergandeng dalam simulasi biosensor responsif-rangsangan.

ABSTRACT

This project addresses the critical implementation gap in computational electromagnetics by developing an open-source finite element solver for light scattering simulation using high-order Nédélec edge elements within the FEniCSx framework. The primary objective is to establish a mathematically rigorous and physically accurate computational tool that democratizes access to electromagnetic scattering analysis while maintaining professional-grade accuracy for applications in biosensor design and nanophotonics research. The research significance lies in bridging the divide between prohibitively expensive commercial software and technically demanding custom implementations, providing the scientific community with a transparent, verifiable, and extensible solver that combines theoretical rigor with practical accessibility. The methodology employs third-order $H(\mathrm{curl})$-conforming Nédélec elements on unstructured triangular meshes to discretize the time-harmonic vector Helmholtz equation, with Perfect Electric Conductor boundary conditions enforced through strong Dirichlet constraints and domain truncation achieved via first-order absorbing boundary conditions. The weak formulation is implemented using the Unified Form Language and solved through direct LU factorization via the MUMPS solver. Validation is performed against analytical Mie series benchmarks for two-dimensional cylindrical scattering in the resonance regime ($ka \approx 3.14$), with systematic convergence studies conducted across five progressively refined meshes. Preliminary results demonstrate successful implementation of the mathematical framework, with qualitative field visualizations confirming physically correct standing wave patterns, geometric shadow formation, and proper boundary condition enforcement without spurious artifacts. Quantitative validation reveals that the solver operates within the stable asymptotic regime ($kh = 0.374 < 0.5$), avoiding numerical pollution effects. The convergence study provides definitive proof of implementation correctness, demonstrating that the numerical error decreases monotonically with mesh refinement and achieves an empirical convergence rate of 2.88, in excellent agreement with the theoretical prediction of 3.0 for cubic Nédélec elements. This close match between observed and theoretical convergence rates confirms that the weak formulation is correctly implemented, boundary conditions are properly enforced, and the solution has entered the asymptotic regime where discretization error dominates over phase error. The validated two-dimensional solver establishes a robust foundation for future extension to three-dimensional geometries and coupled multi-physics applications in stimuli-responsive biosensor simulation.

Table of Contents

List of Tables

List of Figures

LIST OF SYMBOLS AND ABBREVIATIONS

Symbols

E	Electric field intensity
H	Magnetic field intensity
D	Electric flux density
B	Magnetic flux density
J	Conductive electric current density
ρ	Volume electric charge density
ε	Permittivity
μ	Permeability
σ	Conductivity
ε_r	Relative permittivity
μ_r	Relative permeability
k_0	Free space wavenumber
ω	Angular frequency
λ	Wavelength
h	Mesh size / average element size
kh	Dimensionless resolution parameter
ka	Size parameter (resonance regime)
p	Polynomial degree/order of basis functions
Ω	Computational domain/volume
$\partial\Omega$	Domain boundary
$\hat{\boldsymbol{n}}$	Unit normal vector

Abbreviations

ABC	Absorbing Boundary Condition
BEM	Boundary Element Method
FDTD	Finite Difference Time Domain
FDFD	Finite Difference Frequency Domain
FEM	Finite Element Method
HPC	High-Performance Computing
LOD	Limit of Detection
LU	Lower-Upper (Factorization/Decomposition)
MUMPS	MUltifrontal Massively Parallel sparse direct Solver
PEC	Perfect Electric Conductor
PEGDA	Poly(ethylene glycol) diacrylate
PETSc	Portable, Extensible Toolkit for Scientific Computation
PMC	Perfect Magnetic Conductor
PML	Perfectly Matched Layer
TE	Transverse Electric (mode)
UFL	Unified Form Language

Chapter 1

Introduction

1.1 Introduction to Light Scattering

Light scattering is a ubiquitous and fundamental physical phenomenon that describes the interaction of electromagnetic waves with discrete particles or inhomogeneous media. Whether it is the blue color of the sky, the brilliance of a white cloud, or the operation of radar systems, scattering occurs whenever an incident electromagnetic wave encounters an obstacle. This interaction induces oscillating multipoles, described as microscopic electric charge separations that vary with time within the object, which subsequently radiate secondary waves known as the scattered field. Understanding how the amplitude, phase and polarization of light evolve during this interaction is essential for a wide array of scientific and engineering disciplines.

The detailed study of these interactions is a cornerstone for the development of advanced technologies. In remote sensing, the analysis of scattered waves allows for the detection and characterization of objects that are otherwise inaccessible. Similarly, in biomedical optics, the scattering properties of tissues are exploited to diagnose diseases at an early stage. However, the complexity of practical scatterers, which often possess irregular shapes and inhomogeneous material properties, renders analytical solutions such as the Mie series insufficient (Wriedt, 2012). While Mie theory provides an exact mathematical solution for scattering by simple spheres and has served as the gold standard for validation in the field, it cannot be applied to the arbitrary geometries encountered in practical engineering problems. Consequently, numerical simulation has become an indispensable tool. By discretizing the governing equations of electromagnetism and converting the continuous differential equations into a system of algebraic equations solvable by computer, numerical methods allow researchers to predict scattering behavior with high fidelity (Rumpf et al., 2014).

The physics of this interaction is governed entirely by the laws of classical electromagnetism. At the macroscopic level, the behavior of these electromagnetic fields is described by a set of coupled partial differential equations known as Maxwell's equations.

1.1.1 Maxwell's Equations

In the context of light scattering, the fundamental behavior of electromagnetic fields is described by Maxwell's equations. For a linear, isotropic medium characterized by permittivity ε (a measure of electric polarization), permeability μ (a measure of magnetic response) and conductivity σ, the general time-domain differential forms of these equations are:

$$\nabla \times \boldsymbol{E} = -\frac{\partial \boldsymbol{B}}{\partial t} \tag{1.1}$$

$$\nabla \times \boldsymbol{H} = \frac{\partial \boldsymbol{D}}{\partial t} + \boldsymbol{J} \tag{1.2}$$

$$\nabla \cdot \boldsymbol{D} = \rho \tag{1.3}$$

$$\nabla \cdot \boldsymbol{B} = 0 \tag{1.4}$$

where $\boldsymbol{D} = \varepsilon \boldsymbol{E}$ is the electric flux density and $\boldsymbol{B} = \mu \boldsymbol{H}$ is the magnetic flux density.

Faraday's Law of Induction (1.1)**.**

This equation describes the fundamental principle of electromagnetic induction. It states that a time-varying magnetic flux density ($\boldsymbol{B}$) induces a spatially circulating electric field intensity ($\boldsymbol{E}$) (Jin, 2015). Physically, this coupling implies that a changing magnetic environment is capable of generating an electromotive force, which is the mechanism governing devices like transformers and generators (Jin, 2015).

Ampère's Circuital Law (1.2)**.**

This law establishes that the curl of the magnetic field intensity ($\boldsymbol{H}$) is generated by two distinct sources which are the flow of conductive electric current density ($\boldsymbol{J}$) and the time rate of change of the electric flux density ($\boldsymbol{D}$) (Jin, 2015). The latter term, $\frac{\partial \boldsymbol{D}}{\partial t}$, represents the "displacement current," which is critical for explaining how magnetic fields can exist in regions without physical charge flow (such as vacuum or dielectrics), thereby allowing for the propagation of electromagnetic waves (Jin, 2015).

Gauss's Law for Electric Fields (1.3)**.**

Gauss's law for electric fields states that the divergence of the electric flux density ($\boldsymbol{D}$) is directly proportional to the volume electric charge density (ρ) (Jin, 2015). Physically, this signifies that electric charges act as the sources (positive charge) or sinks (negative charge) of the electric field where the flux lines diverge outwards from positive charge and converge inwards towards negative charge (Jin, 2015).

Gauss's Law for Magnetic Fields (1.4)**.**

This equation asserts that the divergence of the magnetic flux density ($\boldsymbol{B}$) is always zero (Jin, 2015). Physically, this implies the non-existence of isolated magnetic charges (magnetic monopoles). Consequently, magnetic field lines are solenoidal, meaning they always form continuous, closed loops with no distinct beginning or end (Jin, 2015).

1.1.2 Significance of Light Scattering Simulation

The simulation of light scattering has become indispensable in biomedical engineering driven by the critical need for non-invasive diagnostic tools and the ethical constraints associated with direct human or animal experimentation (Periyasamy and Pramanik, 2017; Krasnikov et al., 2019). Recognized as the gold standard for modeling light transport, Monte Carlo (MC) simulations allow researchers to rigorously study photon interactions within complex, heterogeneous biological structures which ranging from the intricate layers of the human eye to whole-body small animal models where traditional analytical solutions often fail (Periyasamy and Pramanik, 2017; Krasnikov et al., 2019). The implementation of these methods has been revolutionized by the integration of realistic mesh-based geometries and parallel processing technologies like Graphics Processing Units (GPUs), which address the massive computational demands required to trace millions of photon paths for statistical accuracy (Periyasamy and Pramanik, 2017; Krasnikov et al., 2019). Consequently, these advanced simulations provide a robust platform for optimizing novel imaging systems, such as Optical Coherence Tomography (OCT) and Raman spectroscopy and are essential for quantifying light distribution to ensure safety in therapeutic applications like optogenetics without the risks of invasive procedures (Krasnikov et al., 2019).

Parallel to these advancements in the biomedical domain, light scattering simulation stands as a foundational pillar in the rapidly evolving field of nanophotonics, where modern systems involve intricate light-matter interaction regimes that extend well beyond the diffraction limit (Gallinet et al., 2015). As research progresses from simple plasmonic nanostructures to increasingly sophisticated multiscale devices, traditional analytical methods become insufficient, rendering numerical modeling central to the understanding, control and design of these complex systems (Gallinet et al., 2015). Such simulations are particularly indispensable for engineering exotic metamaterials including sub-wavelength focusing lenses and for optimizing photonic bandgap structures, where the precise manipulation of light propagation, localization and scattering is critical for performance (Teixeira et al., 2023). Consequently, the continuous demand for highly efficient numerical tools is driven by the necessity to accurately model these elaborate physical phenomena, thereby bridging the gap between theoretical nanophotonic concepts and the engineering of functional optical devices for applications ranging from sensing and light harvesting to optical signal processing (Gallinet et al., 2015; Teixeira et al., 2023).

The development of advanced biosensing platforms is driven by the critical clinical need for non-invasive, continuous glucose monitoring to manage the rising prevalence of diabetes. Hu et al. (2023) highlight that conventional finger-prick methods are painful, intermittent, and prone to patient non-compliance, while enzymatic sensors often suffer from environmental instability and poor longevity. To address these limitations, the Poly(ethylene glycol) diacrylate (PEGDA) based biosensor offers a robust alternative, operating on a chemo-mechanical transduction principle where phenylboronic acid (PBA) motifs induce a volumetric phase transition upon glucose binding (Hu et al., 2023; Cai et al., 2021). However, the mechanism of this sensor relies on a continuously deforming domain rather than a static geometry, creating a complex coupled physics scenario where mechanical deformation and optical properties are intrinsically linked. This specific application serves as a compelling archetype to demonstrate the indispensability of light scattering simulation, which transcends simple visualization to provide a fundamental method for interpreting the light-matter interactions that drive next-generation stimuli-responsive diagnostics.

The necessity of this simulation is further elucidated by the requirement to resolve the time-dependent anisotropic deformation inherent to laser-fabricated polymeric architectures. In the context of the PEGDA biosensor, Ennis et al. (2023) observe that microstructures fabricated via Two-Photon Polymerization (2PP) do not swell uniformly but exhibit complex warping and buckling constrained by substrate adhesion. This physical reality renders analytical models like Mie theory insufficient, as they cannot represent such distorted, low-contrast boundaries (Ennis et al., 2023). Light scattering simulation becomes the essential interpretative engine to map these realistic deformation fields into the optical domain. It provides the unique capability to decouple competing variables which is isolating the effects of geometric expansion from refractive index reduction—thereby yielding mechanistic insights that are otherwise inaccessible through experimental characterization alone.

Finally, this biosensor example highlights the transformative impact of simulation in enabling predictive engineering and rigorous performance benchmarking. Adopting the framework established for photonic crystal sensors, the simulation allows researchers to generate computational calibration curves that link mechanical changes to precise optical shifts, predicting critical metrics such as Limit of Detection (LOD) and Sensitivity. This facilitates computational design optimization, where nanostructural parameters are optimized in silico to maximize electromagnetic field confinement prior to physical fabrication, a strategy validated in the development of SERS substrates. By successfully replicating the field enhancements theoretically expected in these complex geometries, the simulation establishes itself as a rigorous, physically accurate methodology essential for designing and validating elastomeric devices with clinical precision (Kumar and Sen, 2025).

1.2 Problem Statement

The precise simulation of light scattering is a cornerstone in the development of advanced nanophotonic and biomedical devices, where functionality often hinges on the interaction between electromagnetic waves and complex, sub wavelength structures. Historically, analytical techniques like the Mie series have served as the primary tool for such analysis (Wriedt, 2012). However, as detailed by Martin (1993), the applicability of these formulae is rigorously constrained to perfect spheres or infinite cylinders. This introduces significant errors not only when applied to the anisotropic, swelling microstructures found in hydrogel biosensors but also when modeling irregular biological organelles or non-canonical nanostructures. To address these geometric limitations, numerical partial differential equation solvers are utilized, though each approach presents distinct trade-offs. The Finite Difference Time Domain (FDTD) method is often cited for its versatility; Drezek et al. (2000) demonstrate its utility in modeling light scattering from heterogeneous biological cells, where it offers greater geometric flexibility than Mie theory. Furthermore, Teixeira et al. (2023) emphasize FDTD's robustness in handling broad spatial scales and multiphysics scenarios. However, both FDTD and the Finite Difference Frequency Domain (FDFD) method rely on orthogonal Cartesian grids. As Shin (2013) highlights, this discretization inherently suffers from "staircasing" errors when modeling curved interfaces, potentially leading to inaccurate near field predictions for plasmonic nanostructures or deforming sensor surfaces. Similarly, while Boundary Element Methods (BEM) provide rigorous solutions for radiation problems (Cools et al., 2011), they can become computationally expensive when applied to penetrable objects with highly inhomogeneous material properties, such as biological tissues or gradient index lenses (Buffa and Hiptmair, 2003). In this context, the Finite Element Method (FEM) offers a distinct advantage for modeling these complex, curved boundaries through the use of unstructured tetrahedral meshes, which naturally conform to both the deforming geometry of biosensors and the intricate morphology of cellular structures.

The choice between time domain and frequency domain formulations depends heavily on the specific optical setup and the nature of the excitation source. For applications requiring broadband spectral analysis such as analyzing the scattering cross section of a nanoparticle over the entire visible spectrum time domain methods like finite element time domain method are advantageous as they resolve the frequency response in a single simulation run (Teixeira et al., 2023). However, for specific classes of devices interrogated by monochromatic light, the frequency domain numerical approach becomes computationally more efficient. This is particularly relevant for the stimuli responsive biosensors discussed in this study, as well as for designing specialized photonic crystals and metamaterials operating at fixed laser wavelengths. Using a time domain numerical solver for such steady state, monochromatic problems require simulating wave propagation over thousands of time steps until convergence is achieved, a process Drezek et al. (2000) note can be resource intensive. In contrast, a frequency domain numerical solver provides a direct, one-shot solution to the time harmonic Maxwell's equations, matching the continuous wave nature of the excitation. To ensure accuracy in this regime, however, the solver must be carefully designed to avoid artificial errors. Standard nodal based finite element methods often can violate the continuity of tangential electric fields across dielectric boundaries, leading to spurious solutions. Schwarzbach (2009) demonstrates that this flaw frequently leads to spurious solutions that look like real signals but are numerical noise. To prevent this, it is essential to use high order Nédélec (edge) elements, which enforce the correct physical continuity of the electric field, thereby distinguishing true electromagnetic effects (Koba and Szczepanski, 2012) from simulation errors.

Finally, while the theoretical advantages of the finite element method are clear, a significant practical barrier remains due to the lack of a dedicated open source solver specifically architected for electromagnetic scattering. As highlighted by Owusu-Agyemang and Yowetu (2025), the scalable solution of Maxwell's equations in complex 3D geometries faces persistent challenges ranging from prohibitive computational costs to ill conditioned system matrices that require advanced interventions like parallel preconditioners to resolve. Consequently, the main problem aimed to be solved by this project is the prohibitive development curve faced by researchers attempting to utilize raw finite element frameworks, where they must manually formulate the variational weak forms and implement complex high order Nédélec (edge) elements to suppress spurious modes. This manual implementation is historically prone to error because, as established by Nedelec (1980), these vector elements require intricate covariant Piola transformations to map the tangential continuity requirements from a reference element to the distorted physical mesh. To address this bottleneck, this project leverages the FEniCS computing platform, which Otto et al. (2012) demonstrate offers a transformative advantage through its Unified Form Language (UFL). This high level abstraction allows the governing time harmonic Maxwell's equations to be expressed directly as symbolic variational forms, enabling the software to automatically handle the rigorous assembly of edge elements and their associated coordinate mappings. By building upon this advanced architecture, the project bypasses the manual coding errors identified by Boyse et al. (1992) and Jiang et al. (1996) as the primary source of numerical instability. This capability effectively removes the barrier of numerical implementation, empowering researchers to shift their focus from the tedious debugging of solver routines to the creative design and physical analysis required for next generation nanophotonic and biomedical devices.

1.3 Research Objectives

The overarching goal of this research is to develop and validate a robust finite element solver for light scattering simulations. The specific objectives are:

- **Mathematical Formulation and Approximation:** To formulate the variational form of the time harmonic equation.
- **Solver Development and Stabilization:** To implement a high-accuracy finite element solver utilising unstructured meshes and vector edge elements (Nédélec elements) (Nedelec, 1980) to ensure physical correctness and geometric flexibility.
- **Domain Truncation and Validation:** To validate robust domain truncation techniques, specifically absorbing boundary conditions (ABC), to simulate an unbounded free-space environment within a finite computational domain.

1.4 Scope of Work

This project focuses on the development of a finite element solver for Maxwell's equations in linear, isotropic, and non-magnetic media. The analysis is strictly confined to the frequency domain, assuming a time-harmonic dependence ($e^{j\omega t}$) for all fields. This approach is chosen to directly compute the steady-state scattering response without the need for time-stepping algorithms. To rigorously handle the open boundary problem, the solver utilizes the Scattered Field Formulation, which decomposes the total field into a known analytic incident wave and a numerically computed scattered wave. This specific formulation establishes the theoretical boundary for the project, necessitating a rigorous examination of the variational weak forms required to maintain the ellipticity of the system and ensure the uniqueness of the solution as discussed by Jiang et al. (1996) and Boyse et al. (1992).

The geometric scope includes the simulation of scattering from dielectric and metallic cylinders with arbitrary 2D cross-sections. However, the discretization methodology is strictly defined by the requirements of high-frequency stability. Guided by the error analysis of Melenk and Sauter (2024), this project prioritizes High-Order (arbitrary degree) Nédélec edge elements on unstructured triangular meshes. This choice defines the scope of the numerical investigation, shifting the focus from traditional h-refinement strategies to p-refinement techniques necessary to suppress the pollution effect inherent to indefinite Helmholtz problems. Furthermore, the reliance on $H(\mathrm{curl})$-conforming elements to enforce tangential continuity implies a strict exclusion of nodal-based methods, which Nedelec (1980) identifies as the primary source of spurious modes in electromagnetic analysis.

Computationally, the project involves assembling global stiffness and mass matrices to form a linear system of equations ($Ax = b$). Because the frequency-domain Helmholtz equation yields an indefinite and complex-symmetric system, the scope of the linear algebra implementation is limited to sparse direct solvers (specifically LU decomposition). As highlighted by Owusu-Agyemang and Yowetu (2025), this system characteristic explicitly excludes standard iterative methods (like Conjugate Gradient) which are generally unstable for wave propagation problems. Consequently, the computational framework is built upon the FEniCS platform to leverage its automated form compilation and interface with robust direct factorization libraries, as validated by Otto et al. (2012).

Validation of the solver is strictly limited to the optical and microwave frequency regimes, utilizing the Analytical Mie Series for a dielectric cylinder as the sole benchmark for verification. As established by Martin (1993), this exact closed-form solution serves as the absolute reference point for verifying solver fidelity. Consequently, the validation scope focuses exclusively on the mathematical derivation of cylindrical harmonics to establish precise error metrics for scattering cross-sections and near-field intensity. Alternative semi-analytical approaches, such as the T-Matrix method, are explicitly excluded from this study to ensure that all numerical results are benchmarked against an exact, analytical standard. Nonlinear optical effects, time-varying material properties, and transient time-domain phenomena are similarly excluded.

Chapter 2

Literature Review

2.1 Development of Mie Series and its Role in Analytical Solutions

The theoretical bedrock of electromagnetic scattering is the Mie Series which provides an exact closed form solution to Maxwell's equations for canonical particles by expanding the electromagnetic fields into infinite sums of vector spherical or cylindrical harmonics (Martin, 1993). Although Martin (1993) notes that this analytical approach is not a versatile design tool for arbitrary geometries due to its strict requirement that object boundaries must coincide perfectly with the coordinate surfaces of an orthogonal system it remains the absolute reference point for accuracy because it eliminates the discretization and staircasing errors inherent in numerical approximations (Wriedt, 2012). Consequently in modern computational electromagnetics the comparison of numerical results against these analytical Mie solutions serves as the primary validation method to verify solver accuracy and ensure the absence of spurious modes before a solver is applied to geometrically complex bodies (Paulsen, 1994). To rigorously bridge the gap between analytical theory and full three dimensional numerical implementation Sjöberg (2025) advocates for the use of rotationally symmetric solvers as a critical validation stage. Sjöberg (2025) theoretically justifies this approach by demonstrating that for bodies of revolution the three dimensional vector wave equation can be decomposed into a sequence of decoupled two dimensional scalar problems via azimuthal mode expansion which significantly reduces computational cost while maintaining mathematical exactness. This allows the underlying finite element machinery including the weak form assembly and boundary condition enforcement to be verified against Mie theory with extremely high mesh fidelity before transitioning to computationally expensive arbitrary 3D geometries (Sjöberg, 2025). In the specific context of this project the Mie series serves as this analytical benchmark for the scattering of a Perfect Electric Conductor PEC cylinder because the separation of variables in cylindrical coordinates allows for the rigorous derivation of exact scattering coefficients by forcing the tangential electric field to vanish at the conducting surface which provides a direct spectral test for the developed solver.

2.2 Comparative Analysis of Numerical Approaches

The selection between time domain and frequency domain numerical approaches is primarily dictated by the spectral characteristics of the optical problem and the computational efficiency required for steady state analysis. While time domain methods are inherently advantageous for broadband simulations where a single transient pulse can recover the spectral response over a wide frequency range (Teixeira et al., 2023) Miller (1994) argues that this efficiency is strictly inverted for resonant or monochromatic systems because the time domain approach relies on determining the late time response where the signal must be tracked for an extended duration until the transient energy fully dissipates. This makes time stepping algorithmically inefficient for high quality factor structures where the fields ring for thousands of periods compared to frequency domain solvers which directly compute the asymptotic steady state solution without iterating through the entire temporal history (Miller, 1994). Furthermore Anees and Angermann (2019) emphasize that the mathematical formulation of time domain finite element methods requires rigorous discretization in both space and time using complex integration schemes such as backward Euler or symplectic algorithms to ensure numerical stability. This creates a significant computational burden because the accuracy of the temporal evolution is strictly coupled to the spatial mesh density meaning that the fine unstructured meshes required to resolve nanophotonic features necessitate infinitesimally small time steps to avoid instability or excessive dispersion error (Anees and Angermann, 2019). Consequently for the analysis of specific photonic devices operating at fixed laser wavelengths the frequency domain approach is superior as it transforms the hyperbolic wave equation into an elliptic boundary value problem that yields the exact time harmonic solution in a single algebraic step thereby avoiding the cumulative dispersion errors and heavy iteration costs associated with long duration time stepping (Shin and Fan, 2012).

The finite-difference frequency-domain (FDFD) method provides a direct algebraic approach to electromagnetic modeling by discretizing the frequency-domain Maxwell's equations, specifically the Helmholtz equation, onto a structured grid (Rumpf et al., 2014). Its implementation typically utilizes the standard Yee cell lattice, where electric and magnetic field components are spatially staggered, allowing differential operators to be approximated by simple finite differences (Rumpf et al., 2014). This formulation assembles the simulation into a large linear system ($Ax = b$) that can be solved directly for the steady-state field distribution (Shin and Fan, 2012). The primary benefit of FDFD lies in its algorithmic simplicity and its flexibility in modeling general anisotropic materials by averaging tensor properties across grid cells (Rumpf et al., 2014). However, a significant limitation is its reliance on rectangular Cartesian grids, which introduces "staircasing" discretization errors when modeling curved boundaries; this can degrade accuracy for complex non-orthogonal geometries unless computationally expensive fine meshes are employed (Shin and Fan, 2012).

In contrast to volumetric approaches, the Boundary Element Method (BEM) reduces the dimensionality of the problem by transforming Maxwell's equations into boundary integral equations using Green's functions (Buffa and Hiptmair, 2003). Instead of discretizing the entire simulation volume, BEM discretizes only the surface of the scatterer into patches, solving for equivalent surface currents rather than volumetric fields (Buffa and Hiptmair, 2003). A major advantage of this implementation is that the underlying Green's functions naturally satisfy the Silver-Müller radiation condition, ensuring exact modeling of wave propagation in unbounded domains without the need for artificial absorbing boundaries (Buffa and Hiptmair, 2003). While this ensures high accuracy for homogeneous scatterers, the method produces fully populated (dense) system matrices because every boundary element interacts with every other element, rendering it computationally inefficient for problems involving highly inhomogeneous or non-linear media compared to sparse methods (Buffa and Hiptmair, 2003).

The finite element method (FEM) is a numerical approach that solves the variational form of the vector wave equation by discretizing the computational domain into an unstructured mesh (Jin, 2015). Unlike structured grids, this unstructured discretization allows for the precise modeling of curved boundaries without geometric approximation errors and naturally accommodates continuously varying material inhomogeneities by defining properties on an element-by-element basis (Paulsen, 1994). However, the method is subject to inherent stability concerns, particularly the accumulation of numerical dispersion errors over large electrical distances which is known as the pollution effect, so it necessitates high mesh densities for high-frequency simulations (Babuška and Sauter, 1997). Furthermore, the formulation requires careful implementation to avoid spurious non-physical solutions and necessitates truncating the unbounded simulation domain, which increases the computational problem size compared to boundary-only methods (Jin, 2015). Despite these challenges, the sparsity of the resulting system matrices makes FEM a highly effective tool for analysing complex, heterogeneous structures.

2.3 Basis Functions and Spurious Modes

The selection of basis functions serves as the fundamental component that dictates the physical fidelity of the finite element method in electromagnetic analysis. This choice is not merely numerical but is governed by the strict continuity conditions imposed by Maxwell's equations at the interfaces between different media. In physical reality, the tangential components of the electric field must remain continuous across material boundaries to satisfy Faraday's law, whereas the normal component must undergo a discrete jump proportional to the discontinuity in the electric permittivity. Bossavit and Mayergoyz (1989) provide a critical justification for rejecting standard node based vectorial finite elements, arguing that they impose continuity on all spatial components of the field across element boundaries. This imposition is a constraint that lacks physical necessity and actively prevents the numerical model from representing the true behavior of the field at dielectric interfaces. When a node-based basis function enforces full continuity in all directions, it artificially smooths the solution, leading to significant errors near boundaries where material properties change abruptly (Bossavit and Mayergoyz, 1989).

To resolve this conflict between numerical formulation and physical law, the focus shifts to vector basis functions, commonly referred to as edge elements. Unlike their nodal counterparts which associate degrees of freedom with the mesh vertices, these elements assign degrees of freedom to the edges of the computational grid. Bossavit and Mayergoyz (1989) demonstrate that this geometric association allows the basis functions to define a space where only the tangential continuity is enforced along the element interfaces. This leaves the normal component free to be discontinuous, thereby aligning the mathematical properties of the discretization with the continuity requirements of the electric field (Kamireddy et al., 2022). By ensuring that the numerical subspace reflects the physical space of the electromagnetic field, edge elements allow for the direct modeling of complex internal inhomogeneities without the need for penalty terms or ad hoc smoothing techniques.

Beyond the issues of interface continuity, the numerical scheme must address the critical issue of spurious modes. These are defined as unphysical solutions that arise within the computational domain, often manifesting as highly divergent fields that pollute the solution spectrum. The origin of these modes lies in the inability of standard nodal elements to properly represent the null space of the curl operator. Bossavit and Mayergoyz (1989) highlight that edge elements possess the distinct advantage of not generating these spurious modes when applied to resonant cavity problems or scattering formulations. This stability arises because edge elements mimic the exact sequence property of the de Rham complex, ensuring that the gradient of the discrete scalar space is contained exactly within the range of the vector edge element space (Schwarzbach, 2009). Consequently, the discrete curl operator mimics the behavior of the continuous operator by mapping gradient fields to exactly zero, effectively separating the unphysical irrotational modes from the physical solenoidal ones.

While nodal elements combined with potential formulations offer a valid pathway by solving for continuous potentials rather than discontinuous fields (Paulsen, 1994), the direct modeling of the electric field using edge elements provides a more robust alignment with the vector nature of Maxwell's equations. This theoretical foundation directly informs the practical development of the finite element solver, which utilizes the open-source platform FEniCSx. The choice of FEniCSx is strategic, as the platform provides a well-established implementation of high order Nédélec elements that are rigorous and mathematically verified. By leveraging these built in capabilities, the solver automatically satisfies the tangential continuity conditions required at material interfaces and possesses the necessary mathematical structure to eliminate spurious modes (Esterhazy, 2013). This integration ensures that the simulation tool achieves both physical accuracy and spectral stability in the analysis of arbitrary electromagnetic structures, as strongly justified by the foundational work of Bossavit and Mayergoyz (1989) and subsequent comparative studies.

2.4 Boundary Conditions and Domain Truncation

The definition of boundary conditions constitutes a pivotal step in the design of a finite element solver, as these constraints serve as the mathematical interface between the finite computational domain and the infinite physical reality of light scattering. The most immediate challenge in scattering simulations is the truncation of the unbounded domain. Since the scattered field propagates outward indefinitely, the numerical model must satisfy the Sommerfeld radiation condition, which requires that the energy radiates away from the source without reflection at infinity (Schwarzbach, 2009). To mimic this behavior within a finite mesh, early development stages in this project will prioritize Absorbing Boundary Conditions (ABCs). These apply local differential operators to the boundary surface to approximate the one-way wave equation (Engquist and Majda, 1977). This choice is driven by the ease of implementation, as ABCs can be integrated directly into the surface integrals of the weak form without requiring the construction of auxiliary geometric domains. However, while computationally inexpensive, these local operators frequently introduce spurious reflections when electromagnetic waves strike the boundary at oblique angles, thereby degrading the accuracy of the solution in the near-field region.

As the solver development progresses towards higher precision requirements, the focus inevitably shifts from ABCs to the Perfectly Matched Layer (PML). As established by Berenger (1994), the PML is not a simple boundary condition but rather an artificial absorbing volume surrounding the physical domain. By utilizing complex coordinate stretching or anisotropic material tensors, the PML promotes the exponential decay of electromagnetic waves inside the layer without generating reflections at the interface between the free space and the absorbing medium. Grand and Le Ru (2020) provide critical practical justification for this transition, demonstrating that while PMLs are theoretically reflectionless, their implementation requires a sufficiently dense mesh, typically at least five elements across the layer thickness to prevent numerical discretization error. This approach ensures that the solver faithfully emulates an open boundary, allowing for the accurate retrieval of scattering cross-sections and far-field patterns, effectively eliminating the reflection artifacts inherent to simpler ABC implementations.

Beyond the truncation of exterior boundaries, the imposition of internal boundary conditions is essential for ensuring physical fidelity and exploiting geometric symmetries to reduce computational cost. Chaber et al. (2017) discuss the implementation of Perfect Electric Conductor (PEC) and Perfect Magnetic Conductor (PMC) conditions within the finite element framework. The PEC condition, which forces the tangential electric field to zero, is the standard mathematical representation for highly conductive metals where fields do not penetrate. Conversely, the PMC condition forces the tangential magnetic field to zero. While PMCs do not exist as natural materials, Chaber et al. (2017) highlight their utility in imposing symmetry planes. By analyzing the incident field polarization, a scattering problem on a symmetric object can often be reduced to one-half or one-quarter of the original domain by applying appropriate PEC or PMC conditions on the cut planes, significantly reducing the memory footprint and solution time.

From the perspective of mathematical stability and formulation, the choice of boundary condition dictates the well-posedness of the system matrix. In the weak variational form used by solvers like FEniCSx, boundary conditions are categorized as either essential (Dirichlet) or natural (Neumann). Schwarzbach (2009) argues that the stability of the finite element solution, particularly at low frequencies, relies on the correct application of these conditions to prevent the singularity of the system matrix. For instance, when solving the curl-curl equation, the lack of proper boundary constraints on the gradient subspace can lead to an ill-conditioned system. By rigorously defining the essential boundary conditions on the electric field (such as PEC on metal surfaces) and allowing the natural boundary conditions to handle the magnetic field continuity via surface integrals, the solver ensures a unique and stable solution. This interplay between physical constraints and mathematical formulation confirms that boundary conditions are not merely auxiliary settings but are intrinsic to the validity of the electromagnetic simulation.

2.5 Error Analysis and the Pollution Effect

The implementation of error analysis within a finite element solver for electromagnetic analysis is not merely a validation step but a theoretical necessity required to certify the physical fidelity of the simulation. In the context of solving time harmonic Maxwell equations, the importance of this analysis stems from the inherent difficulty in distinguishing between a physically converged solution and a numerical artifact caused by discretization errors. Melenk and Sauter (2024) emphasize that standard error estimation techniques often fail in high frequency regimes due to the pollution effect, where the numerical phase error accumulates as the wave propagates through the mesh. This accumulation leads to a global dispersion error that grows with the wavenumber, meaning that a locally accurate solution can be globally incorrect. Therefore, the primary function of error analysis is to quantify this deviation using specific mathematical norms, ensuring that the chosen discretization strategy possesses the necessary stability to suppress these pollution errors and yield a reliable approximation of the electromagnetic field.

To perform a rigorous error analysis, one must move beyond simple visual inspection and utilize quantitative metrics established in the mathematical literature. Zhong et al. (2018) provide the theoretical framework for optimal error estimates by utilizing the $H(\text{curl})$ norm alongside the standard L^2 norm. The L^2 norm measures the average energy error over the domain, while the $H(\text{curl})$ norm accounts for the accuracy of the field's rotation, which is critical for vector electromagnetics. In scenarios where an analytical solution is available, such as the Mie series for a sphere, the error is calculated directly as the difference between the numerical and exact fields. However, for practical engineering problems involving complex geometries where no analytical solution exists, the convergence analysis operates via a process of self-validation (Esterhazy, 2013). This involves computing a reference solution on an extremely refined mesh, often referred to as an overkill mesh, and treating it as the ground truth. The error for coarser discretisation is then evaluated relative to this reference solution to verify that the results form a Cauchy sequence, meaning the difference between successive refinements approaches zero (Grand and Le Ru, 2020).

The mechanism of convergence analysis in these complex scenarios relies on identifying the asymptotic regime of the solver. Frelet et al. (2024) distinguish between the pre asymptotic regime, where the mesh is too coarse relative to the wavelength causing the error to oscillate unpredictably, and the asymptotic regime, where the error decreases monotonically. By plotting the relative error against the number of degrees of freedom on a logarithmic scale, one can extract the convergence rate. If the solver is functioning correctly, this rate should match the theoretical order of the polynomial basis functions used. Furthermore, for problems lacking a reference solution, one can analyse the residual error, which measures how well the numerical solution satisfies the original partial differential equation locally within each element. Hazard and Lenoir (1996) suggest that comparing the standard curl curl solution against a regularized formulation provides another layer of validation, ensuring that the solution remains stable and free from spurious modes even in the presence of geometric singularities.

The collective insights from these error and convergence analyses provide a definitive directive for the development of the finite element solver which ensure the adoption of high order elements is indispensable. The theoretical work by Melenk and Sauter (2024) explicitly demonstrates that mesh refinement alone, known as h refinement, is computationally inefficient for suppressing the pollution effect at high frequencies. They establish a strict stability condition requiring that the ratio of the wavenumber and mesh size to the polynomial degree remains small. This condition implies that increasing the polynomial degree, or p refinement, is the only robust mechanism to relax the mesh density requirements while maintaining accuracy. Nicaise and Tomezyk (2020) reinforce this by showing that high order Nédélec elements unlock exponential convergence rates, allowing the solver to reach the stable asymptotic regime with significantly fewer degrees of freedom compared to low order methods. Consequently, the error analysis confirms that utilizing arbitrary order edge elements within the FEniCSx platform is the optimal strategy to ensure both spectral stability and computational efficiency in the simulation of light scattering.

2.6 Summary and Research Gaps

The existing literature establishes a comprehensive blueprint for physically accurate electromagnetic solvers by noting that the mathematical framework for two-dimensional frequency-domain finite element methods is well established. Foundational principles such as the use of Nédélec edge elements to ensure tangential field continuity, the application of absorbing boundary conditions for open domain scattering, and rigorous discretization strategies to maintain stability are all extensively documented in current research. However, a critical implementation gap persists that prevents these proven methodologies from being accessible to the broader light scattering research community. This gap manifests across the current landscape of simulation tools by creating significant barriers to entry for many researchers who must choose between prohibitive costs or extreme technical complexity.

The primary challenge in light scattering simulation is not a lack of theoretical knowledge but rather the practical difficulty of accessing or developing reliable computational tools. While established commercial platforms offer powerful tools, their proprietary nature often prevents deep algorithmic verification and limits transparency for independent reproduction. Furthermore, these closed architectures can be difficult to customize for the unique coupled physics scenarios inherent to stimuli-responsive biosensors, such as directly linking mechanical deformation data to optical simulation inputs. Alternatively, raw general-purpose finite element libraries provide high mathematical flexibility but demand deep expertise in computational mathematics and advanced programming. Implementing a complete electromagnetic solver from these first principles requires proficiency in variational formulations, $H(\text{curl})$ function spaces, and complex mesh generation. For biosensor researchers whose primary expertise lies in chemistry, materials science, or experimental optics, these technical requirements represent a significant barrier that often leads to the use of oversimplified analytical models that cannot account for complex deforming geometries. Additionally, a significant portion of electromagnetic research relies on in-house or custom-built codes that are rarely made available to the public and frequently lack the comprehensive documentation, standardized validation suites, and long-term maintenance required for scientific reproducibility. Even when such codes are released, they are often undocumented and require expert-level understanding to adapt, or they may depend on specific proprietary libraries and infrastructure that are not accessible to all researchers.

A critical gap remains in the unification of these advanced methodologies into a single accessible framework. While the literature defines individual components for high-order stability, optimized parallel solvers, and rigorous domain truncation, there is currently no single open-source implementation that efficiently combines high-order edge elements, automated variational assembly, and robust validation protocols specifically tailored for multi-scale scattering problems. This research aims to bridge that gap by integrating these disparate theoretical advancements into a cohesive and verifiable solver that lowers the technical barrier for the light scattering community while maintaining the highest standards of physical and mathematical accuracy.

Chapter 3

Methodology

3.1 Introduction

The primary objective of this methodology is to construct a robust, high-fidelity finite element solver capable of simulating electromagnetic scattering by arbitrary objects. The solver is first implemented for a two-dimensional (2D) scenario—specifically, the scattering of a plane wave by a Perfect Electric Conductor (PEC) cylinder. This simplified case allows for direct validation of the governing PDE, boundary condition implementation and weak form assembly against the rigorous analytical benchmark of the Mie Series. Only after the 2D solver is proven to reproduce the analytical results with high accuracy will the framework be extended to full 3D vector implementations.

3.2 Mathematical Formulation

The derivation of the governing equations begins from the fundamental time domain Maxwell's equations in differential form. For a linear and isotropic medium characterized by position dependent permittivity $\varepsilon(\boldsymbol{r})$ and permeability $\mu(\boldsymbol{r})$ the equations are given by Jin (2015) as

$$\nabla \times \boldsymbol{E}(t) = -\frac{\partial \boldsymbol{B}(t)}{\partial t} \quad \text{and} \quad \nabla \times \boldsymbol{H}(t) = \frac{\partial \boldsymbol{D}(t)}{\partial t} + \boldsymbol{J}(t) \tag{3.1}$$

where the bold notation denotes time dependent field vectors. To facilitate the analysis of steady state light scattering the solver adopts the frequency domain approach by assuming a time harmonic dependence $e^{j\omega t}$ for all fields which transforms the time derivative operator $\frac{\partial}{\partial t}$ into multiplication by $j\omega$ (Monk, 1992). By applying the constitutive relations $\boldsymbol{D} = \varepsilon \boldsymbol{E}$ and $\boldsymbol{B} = \mu \boldsymbol{H}$

and assuming a source free region ($\boldsymbol{J} = 0$) the coupled first order Maxwell's equations become

$$\nabla \times \boldsymbol{E} = -j\omega\mu \boldsymbol{H} \tag{3.2}$$

$$\nabla \times \boldsymbol{H} = j\omega\varepsilon \boldsymbol{E} \tag{3.3}$$

To obtain a single governing equation for the electric field, we decouple the system by taking the curl of Faraday's Law:

$$\nabla \times (\mu^{-1}\nabla \times \boldsymbol{E}) = -j\omega\nabla \times \boldsymbol{H} \tag{3.4}$$

Substituting Ampere's Law ($\nabla \times \boldsymbol{H} = j\omega\varepsilon\boldsymbol{E}$) into the right-hand side gives:

$$\nabla \times (\mu^{-1}\nabla \times \boldsymbol{E}) = -j\omega(j\omega\varepsilon\boldsymbol{E}) = \omega^2\varepsilon\boldsymbol{E} \tag{3.5}$$

Rearranging terms leads to the Vector Wave Equation (Curl-Curl Equation):

$$\nabla \times \left(\frac{1}{\mu_r}\nabla \times \boldsymbol{E}\right) - k_0^2\varepsilon_r\boldsymbol{E} = 0 \tag{3.6}$$

where $k_0 = \omega\sqrt{\mu_0\varepsilon_0}$ represents the free space wavenumber and ε_r is the relative permittivity (Jin, 2015). However solving this equation directly for the total electric field $\boldsymbol{E}_{\text{tot}}$ in an unbounded domain is numerically problematic due to the propagation of the incident wave to infinity. To rigorously handle the open boundary problem and the domain truncation the solver utilizes the Scattered Field Formulation as described by Otto et al. (2012). This approach decomposes the total electric field into the sum of a known analytical incident field $\boldsymbol{E}_{\text{inc}}$ and an unknown scattered field $\boldsymbol{E}_{\text{sc}}$ such that $\boldsymbol{E}_{\text{tot}} = \boldsymbol{E}_{\text{inc}} + \boldsymbol{E}_{\text{sc}}$. By substituting this decomposition into the vector wave equation and noting that the incident field satisfies Maxwell's equations in free space the problem is recast into a form where the scatterer acts as a radiation source. The final strong form governing equation utilized in this project is

$$\nabla \times \left(\frac{1}{\mu_r}\nabla \times \boldsymbol{E}_{\text{sc}}\right) - k_0^2\varepsilon_r\boldsymbol{E}_{\text{sc}} = -\nabla \times \left(\frac{1}{\mu_r}\nabla \times \boldsymbol{E}_{\text{inc}}\right) + k_0^2\varepsilon_r\boldsymbol{E}_{\text{inc}} \tag{3.7}$$

where the right hand side represents the equivalent current source generated by the dielectric contrast between the object and the background medium (Jin, 2015). This formulation is advantageous because it allows the Absorbing Boundary Conditions to be applied solely to the scattered field $\boldsymbol{E}_{\text{sc}}$ which satisfies the Sommerfeld radiation condition at infinity thereby preventing non physical reflections of the incident wave at the truncation boundary (Otto et al., 2012).

3.3 Weak Formulation

To solve the governing partial differential equation using the finite element method, the continuous strong form problem must be projected onto a discrete function space which transforms the second-order differential equation into a variational statement that relaxes the strict continuity requirements of the solution. As derived by Jin (2015), the process initiates by taking the inner product of the vector wave equation with a vector test function $\boldsymbol{v}$ chosen from the Hilbert space $H(\text{curl};\Omega)$, which consists of square-integrable vector functions whose curl is also square-integrable (Anees and Angermann, 2019). Multiplying the scattered field equation by $\boldsymbol{v}$ and integrating over the computational volume Ω yields the weighted residual form

$$\int_{\Omega}\left(\nabla\times\left(\frac{1}{\mu_r}\nabla\times\boldsymbol{E}_{\text{sc}}\right)\right)\cdot\boldsymbol{v}\,d\Omega-\int_{\Omega}k_0^2\varepsilon_r\boldsymbol{E}_{\text{sc}}\cdot\boldsymbol{v}\,d\Omega=\int_{\Omega}\boldsymbol{F}_{\text{inc}}\cdot\boldsymbol{v}\,d\Omega \tag{3.8}$$

where $\boldsymbol{F}_{\text{inc}}$ represents the source term derived from the incident field. To properly handle the second-order curl operator $\nabla\times\nabla\times$ and distribute the differentiation between the trial and test functions, the formulation employs Green's First Vector Identity. This mathematical identity states that for any two vector fields $\boldsymbol{A}$ and $\boldsymbol{B}$, the volume integral of the curl-curl product is related to a symmetric bilinear form and a surface boundary term by the relation

$$\int_{\Omega}(\nabla\times\nabla\times\boldsymbol{A})\cdot\boldsymbol{B}\,d\Omega=\int_{\Omega}(\nabla\times\boldsymbol{A})\cdot(\nabla\times\boldsymbol{B})\,d\Omega-\oint_{\partial\Omega}(\boldsymbol{A}\times\nabla\times\boldsymbol{B})\cdot\hat{\boldsymbol{n}}\,dS \tag{3.9}$$

By applying this identity with $\boldsymbol{A}=\boldsymbol{E}_{\text{sc}}$ and $\boldsymbol{B}=\boldsymbol{v}$, the formulation shifts one curl operator from the unknown electric field onto the test function. Jin (2015) emphasizes that this step is mathematically critical because it reduces the differentiability requirement from C^1 to $H(\text{curl})$ continuity and naturally exposes the surface integral term required for boundary conditions. Substituting this identity into the weighted residual equation yields the final bilinear weak form utilized in the solver

$$\begin{aligned}\int_{\Omega}\left[\frac{1}{\mu_r}(\nabla\times\boldsymbol{E}_{\text{sc}})\cdot(\nabla\times\boldsymbol{v})-k_0^2\varepsilon_r\boldsymbol{E}_{\text{sc}}\cdot\boldsymbol{v}\right]d\Omega&\\+\int_{\partial\Omega}\left(\hat{\boldsymbol{n}}\times\frac{1}{\mu_r}\nabla\times\boldsymbol{E}_{\text{sc}}\right)\cdot\boldsymbol{v}\,dS&=\int_{\Omega}k_0^2(\varepsilon_r-1)\boldsymbol{E}_{\text{inc}}\cdot\boldsymbol{v}\,d\Omega\end{aligned} \tag{3.10}$$

where the surface integral over $\partial\Omega$ represents the tangential magnetic field flux through the domain boundary. Sjöberg (2025) notes that this "natural" boundary term provides the mechanism to rigorously couple the interior domain to the exterior radiation condition (ABC) or the conducting surface (PEC) without auxiliary equations. By solving this weak form, the finite element method ensures a stable, parasite-free solution that rigorously satisfies the energy conservation properties of Maxwell's equations (Boyse et al., 1992).

3.4 Boundary Conditions

The mathematical well-posedness of the finite element model relies strictly on the correct imposition of boundary conditions which serve as the interface between the finite computational domain and the physical reality of the scattering problem (Anees and Angermann, 2019). To accurately simulate the interaction of the incident wave with the metallic cylinder, the solver must first enforce the Perfect Electric Conductor (PEC) condition on the object's surface Γ_{PEC}. Physically, this condition dictates that the total tangential electric field must vanish at the interface of a perfect conductor since no field can exist within the material (Sjöberg, 2025). In the context of the scattered field formulation derived in the previous section, this physical constraint translates into a mathematical requirement that the tangential component of the scattered field must exactly oppose the tangential component of the incident field such that $\hat{\boldsymbol{n}} \times \boldsymbol{E}_{\text{sc}} = -\hat{\boldsymbol{n}} \times \boldsymbol{E}_{\text{inc}}$ on the conducting boundary (Jin, 2015). This is classified as an essential Dirichlet boundary condition because it directly prescribes the value of the unknown field on the boundary rather than its derivative. Consequently, in the finite element assembly, this condition is not added to the weak form integral but is instead enforced algebraically by constraining the system matrix rows corresponding to the boundary degrees of freedom to strictly satisfy this vector equality (Otto et al., 2012).

Simultaneously, the solver must address the truncation of the unbounded exterior domain using an absorbing boundary condition (ABC) applied at the artificial outer boundary Γ_{ABC}. To prevent non-physical reflections from contaminating the solution, the formulation employs the first-order Sommerfeld radiation condition which approximates the behavior of a purely outgoing spherical wave. Mathematically, this is expressed as a Robin-type boundary condition where the curl of the electric field is linearly related to the field itself by $\hat{\boldsymbol{n}} \times \nabla \times \boldsymbol{E}_{\text{sc}} \approx -jk_0\hat{\boldsymbol{n}} \times (\hat{\boldsymbol{n}} \times \boldsymbol{E}_{\text{sc}})$ (Jin, 2015). Unlike the PEC condition, this relationship is a natural Neumann boundary condition which allows it to be substituted directly into the surface integral term of the generic weak formulation. By replacing the generic tangential flux term with this impedance relationship, the final weak form specifically for the scattering problem becomes

$$\int_{\Omega} \left[\frac{1}{\mu_r} (\nabla \times \boldsymbol{E}_{\text{sc}}) \cdot (\nabla \times \boldsymbol{v}) - k_0^2 \varepsilon_r \boldsymbol{E}_{\text{sc}} \cdot \boldsymbol{v} \right] d\Omega + \int_{\Gamma_{\text{ABC}}} jk_0 (\hat{\boldsymbol{n}} \times \boldsymbol{E}_{\text{sc}}) \cdot (\hat{\boldsymbol{n}} \times \boldsymbol{v}) \, dS = \int_{\Omega} k_0^2 (\varepsilon_r - 1) \boldsymbol{E}_{\text{inc}} \cdot \boldsymbol{v} \, d\Omega \quad (3.11)$$

where the surface integral now represents the energy absorption at the domain truncation. The selection of this first-order absorbing boundary condition over more complex perfectly matched layers (PML) is justified by Sjöberg (2025), who argues that for canonical convex scatterers like the PEC cylinder utilized in this validation phase, the first-order approximation provides sufficient accuracy to verify the solver kernel against Mie series benchmarks without introducing the additional numerical stiffness associated with perfectly matched layer implementations. This strategy aligns with the "step-by-step" validation protocol where the mathematical correctness of the weak form assembly is isolated and verified using the simplest robust boundary condition before progressing to complex arbitrary geometries.

3.5 Domain Discretization and Basis Functions

The spatial discretization of the computational domain is achieved using unstructured triangular meshes which offer significant geometric flexibility compared to structured grids thereby allowing for the accurate conformal modeling of curved dielectric interfaces without the staircasing errors inherent in finite difference methods as detailed by Esterhazy (2013). However effective discretization for the time harmonic Helmholtz equation requires rigorously addressing the numerical dispersion error known as the pollution effect which typically degrades solution accuracy as the frequency increases. To mitigate this the solver adopts the wavenumber explicit analysis strategies proposed by Melenk and Sauter (2010) which demonstrate that the numerical error grows with the wavenumber and that high order finite element approximations (p-refinement) are mathematically necessary to maintain stability for high frequency scattering problems.

In conjunction with this discretization strategy the selection of basis functions is critical where the solver explicitly utilizes vector Nédélec edge elements instead of conventional scalar nodal elements. Webb (1999) in his analysis of practical finite element implementations argues that this choice is mandated by the physical requirement that the electric field must maintain tangential continuity across material interfaces while allowing the normal component to be discontinuous which is a condition that standard nodal elements fail to satisfy as they enforce continuity in all spatial directions. Furthermore Monk (1992) emphasizes that for time harmonic scattering problems the application of nodal based finite elements to the vector wave equation is mathematically flawed because it fails to properly represent the null space of the curl operator which inevitably leads to the generation of non physical spurious modes that contaminate the numerical solution. By employing edge elements the formulation inherently satisfies the divergence free condition of the electric field in source free regions and ensures that the gradient of any scalar field maps exactly into the kernel of the curl operator thereby guaranteeing spectral correctness and numerical stability as established by Nedelec (1980).

3.6 The FEniCSx Software for Numerical Implementation

The implementation of the finite element solver is realized using the FEniCSx computing package (specifically the DOLFINx library), which represents the modern, high-performance evolution of the FEniCS project described by Otto et al. (2012) designed to solve partial differential equations (PDEs) through automated code generation. The primary justification for this selection lies in its use of the Unified Form Language (UFL), a domain-specific language that allows the weak form of Maxwell's equations to be expressed in Python code that mirrors the mathematical notation almost exactly. Otto et al. (2012) highlight this as a transformative capability, noting that it allows for the rapid implementation of complex expressions associated with finite element analysis while minimizing the "translation error" between theoretical derivation and code implementation. As demonstrated in their implementation of near-to-far-field transforms, UFL is sufficiently complete to express intricate electromagnetic operations directly as compilable forms, ensuring that the variational problem solved by the computer is strictly identical to the one derived on paper (Otto et al., 2012).

Beyond mathematical transparency, FEniCSx provides crucial support for the automated assembly of vector edge elements. Unlike many general-purpose finite element packages that lack support for vector bases, FEniCS provides native, built-in support for curl-conforming Nédélec elements of the first and second kind (Otto et al., 2012). Otto et al. (2012) emphasize that this native support makes FEniCS an attractive environment for prototyping finite element codes in electromagnetics, as it eliminates the need for the user to manually construct complex vector basis functions or covariant coordinate mappings. This capability is critical for this project as it allows for the effortless implementation of $H(\text{curl})$ conforming spaces, which are required to strictly enforce tangential continuity and prevent the emergence of spurious modes.

Furthermore, the software addresses computational bottlenecks via scalable linear algebra. Owusu-Agyemang and Yowetu (2025) warn that the solution of Maxwell's equations in complex 3D geometries faces persistent challenges, specifically prohibitive computational costs and ill-conditioned system matrices that cause standard iterative solvers to fail. FEniCSx addresses this by interfacing directly with the PETSc (Portable, Extensible Toolkit for Scientific Computation) backend, providing access to advanced parallel preconditioners such as the Auxiliary Maxwell Solver (AMS) or hypre_ams that are essential for stability. This architecture allows the solver to scale across thousands of cores, overcoming the computational limits that Owusu-Agyemang and Yowetu (2025) note are often inaccessible in rigid commercial packages due to their opaque nature.

Finally, this drives the requirement for transparency and reproducibility. Owusu-Agyemang and Yowetu (2025) argue that advancing computational photonics requires moving away from opaque proprietary package to software that prioritize direct mathematical control. By utilizing FEniCSx, this project ensures that the solver logic starting from the weak form definition to the specific matrix assembly kernels to be fully exposed and verifiable, directly addressing the need for scientific reproducibility in modeling complex nanophotonic devices.

3.7 Solver Algorithm Implementation

The computational execution of the finite element solver follows a sequential pipeline that begins with the rigorous discretization of the physical domain. Utilizing the Gmsh generator, the simulation geometry is tessellated into an unstructured grid of triangular elements which naturally conform to the curvature of the cylindrical scatterer. This unstructured discretization is critically selected to eliminate the geometric approximation errors known as "staircasing" which are inherent in structured Cartesian grid methods like FDTD (Shin, 2013). Jin (2015) further emphasizes that for scattering problems involving curved dielectric interfaces, such conformal meshing is essential to preserve the fidelity of the boundary conditions and prevent the generation of spurious numerical reflections that would otherwise degrade the near-field accuracy. During this mesh generation process, physical groups are explicitly tagged to distinguish the topological boundaries:

- the cylinder surface Γ_{PEC} is marked for the imposition of Dirichlet constraints
- the outer domain boundary Γ_{ABC} is identified for the integration of the Absorbing Boundary Condition.

Following the geometric setup, the continuous problem is projected onto a discrete function space V_h constructed using Nédélec Edge Elements of the first kind. To address the numerical pollution effect which degrades accuracy in high-frequency wave propagation problems, the solver employs a high-order polynomial approximation (specifically $p = 3$). This design choice aligns with the stability condition derived by Melenk and Sauter (2024), effectively ensuring that the numerical dispersion error is exponentially suppressed by increasing the interpolation order rather than merely refining the mesh size. The mathematical weak form derived in Section 3.3 is then translated directly into the Unified Form Language (UFL), allowing the FEniCSx form compiler to automatically generate the optimized low-level kernels required to assemble the global stiffness matrix $\boldsymbol{A}$ and the load vector $\boldsymbol{b}$.

The final stage of the algorithm involves solving the resulting linear system $\boldsymbol{Ax} = \boldsymbol{b}$ which is characteristic of the frequency-domain Helmholtz equation: sparse, complex-symmetric, and indefinite. While iterative methods are often employed for massive 3D problems, Owusu-Agyemang and Yowetu (2025) warn that they are prone to convergence stagnation for indefinite wave problems unless specialized preconditioners are used. Consequently, for this 2D validation phase, the solver utilizes a robust sparse direct solver (LU decomposition) provided by the MUMPS library via the PETSc backend. This approach guarantees a stable and accurate solution in a deterministic runtime, isolating the validation focus strictly on the physics of the discretization rather than the tolerances of an iterative solver (Sjöberg, 2025).

3.8 Finite Element Solver Validation

Based on the theoretical frameworks detailed in Wriedt (2012) and Martin (1993), the derivation of the Mie series analytical solution represents a systematic reduction of vector calculus into scalar algebra, transforming the complex interaction of light with a spherical particle into a solvable set of linear equations.

3.8.1 Derivation of the Mie Series

The process begins with the source-free Maxwell's equations. Assuming a time-harmonic field with dependence $e^{-i\omega t}$, these equations are decoupled to form the Vector Helmholtz Equation:

$$\nabla^2 \boldsymbol{E} + k^2 \boldsymbol{E} = 0 \tag{3.12}$$

Solving this vector equation directly in spherical coordinates is mathematically intractable because the unit vectors $(\hat{\boldsymbol{r}}, \hat{\boldsymbol{\theta}}, \hat{\boldsymbol{\phi}})$ are position-dependent. To overcome this, the solution is constructed using Vector Spherical Harmonics (VSH), denoted as $\boldsymbol{M}$ and $\boldsymbol{N}$, which are generated from a scalar potential ψ satisfying the scalar wave equation (Wriedt, 2012):

$$\boldsymbol{M} = \nabla \times (\boldsymbol{r}\psi) \quad \text{and} \quad \boldsymbol{N} = \frac{1}{k} \nabla \times \boldsymbol{M} \tag{3.13}$$

The scalar function ψ is determined via separation of variables, yielding a product of spherical Bessel functions (radial), associated Legendre polynomials (polar), and exponential functions (azimuthal).

Once the basis functions are defined, the electromagnetic fields are expressed as weighted sums of these harmonics:

- **The Incident Field:** Expanded via the Rayleigh expansion into an infinite series of spherical harmonics.
- **The Internal Field:** Expanded using regular Bessel functions (j_n or ψ_n) to ensure finiteness at the sphere's center.

- **The Scattered Field:** Expanded using spherical Hankel functions ($h_n^{(1)}$ or ζ_n) to satisfy the radiation condition at infinity.

The scattered electric field is explicitly written as:

$$\boldsymbol{E}_{\text{sca}} = \sum_{n=1}^{\infty} E_n (ia_n \boldsymbol{N}_{e1n}^{(3)} - b_n \boldsymbol{M}_{o1n}^{(3)}) \tag{3.14}$$

The unknown coefficients a_n and b_n are determined by enforcing the continuity of tangential Electric and Magnetic fields at the sphere's surface ($r = a$). Solving the resulting linear system yields the Mie scattering coefficients:

$$a_n = \frac{m\psi_n(mx)\psi_n'(x) - \psi_n(x)\psi_n'(mx)}{m\psi_n(mx)\xi_n'(x) - \xi_n(x)\psi_n'(mx)} \tag{3.15}$$

$$b_n = \frac{\psi_n(mx)\psi_n'(x) - m\psi_n(x)\psi_n'(mx)}{\psi_n(mx)\xi_n'(x) - m\xi_n(x)\psi_n'(mx)} \tag{3.16}$$

Here, the numerator describes the interference matching the boundary fields, while the denominator dictates the resonance conditions of the sphere (Martin, 1993).

3.8.2 Solver Validation Methodology

Validation begins with a direct visual comparison between the numerically generated field distribution and the analytical benchmark. As described in Wriedt (2012), the total electric field magnitude $|E_{\text{total}}|$ must exhibit specific interference characteristics. In the illuminated region ($x < -a$), a standing wave pattern should emerge with alternating bright and dark fringes, resulting from the interaction between the incident and reflected waves. Conversely, the shadow region ($x > a$) should demonstrate a significant amplitude drop, confirming the blocking effect of the conductor. While this visual inspection provides a qualitative "sanity check," Frelet et al. (2024) warn that qualitative agreement can mask pre-asymptotic errors in high-frequency regimes, necessitating rigorous norm-based quantification.

To rigorously quantify the solver's fidelity, a global error analysis is employed using the relative $L_2(\Omega)$ norm, defined as:

$$\text{Error} = \frac{||\boldsymbol{E}_{\text{num}} - \boldsymbol{E}_{\text{ana}}||_{L^2(\Omega)}}{||\boldsymbol{E}_{\text{ana}}||_{L^2(\Omega)}} \tag{3.17}$$

This metric is selected because it aggregates discrepancies over the entire computational domain rather than at isolated points. Hsiao et al. (2002) emphasize that the L_2 norm provides a stable measure of convergence for finite element schemes involving scattering, as it effectively smooths out local singularities near sharp conducting edges that might otherwise skew L_∞ (maximum norm) estimates. Furthermore, Melenk and Sauter (2024) demonstrate that minimizing this norm directly correlates to the stability of the wave-number explicit solution, ensuring that the numerical solution $\boldsymbol{E}_{\text{num}}$ strictly adheres to the continuity requirements of the underlying Hilbert space $H(\text{curl})$.

The ultimate test of the solver's mathematical correctness is the verification of its convergence rate, which measures how rapidly the numerical error decays as the mesh density increases. In standard finite element theory, refining the mesh size h (h-refinement) is expected to reduce the error according to a specific power law. Frelet et al. (2024) provide the sharpest theoretical bounds for this behavior, proving that for Nédélec edge elements of polynomial order p, the error in the asymptotic regime must satisfy the inequality $\|\boldsymbol{E}-\boldsymbol{E}_h\|_{H(\text{curl})} \leq Ch^p \|\boldsymbol{E}\|_{H^{p+1}}$. However, a critical aspect of validating high-frequency Helmholtz solvers is distinguishing between the asymptotic and pre-asymptotic regimes. Frelet et al. (2024) warn that in the pre-asymptotic phase—where the mesh is sufficiently fine to resolve the geometry but too coarse relative to the wavelength ($kh \not\ll 1$)—the "pollution effect" dominates the solution. In this unstable regime, numerical dispersion errors accumulate as the wave propagates across the domain, causing a significant phase drift between the computed and analytical waves. Consequently, simply refining the mesh in this phase may yield unpredictable results, characterized by error stagnation or even oscillatory behavior where a finer mesh paradoxically produces a higher error due to resonance shifts (Melenk and Sauter, 2024).

The solver is confirmed to have entered the stable asymptotic regime only once the mesh density satisfies the strict resolution threshold ($kh \ll 1$) required to suppress this dispersive accumulation. Once this threshold is crossed, the phase error becomes negligible, and the global error begins to decrease monotonically. To verify this transition, the validation process involves plotting the relative L_2 error against the inverse mesh size $1/h$ on a log-log scale. A linear slope emerging in the tail of this plot that matches the theoretical polynomial order p (e.g., a slope of 3 for cubic elements) serves as the definitive mathematical proof that the implementation is free from "locking" effects and spurious modes, verifying that the discrete weak form has converged to the continuous physical reality as predicted by the stability theory of Hazard and Lenoir (1996).

Chapter 4

Preliminary Results and Discussion

4.1 Introduction

This chapter presents the preliminary results obtained from the finite element solver developed for electromagnetic scattering analysis. The implementation is validated through three complementary approaches: first, an overview of the code structure and computational workflow is provided to establish the methodological foundation; second, qualitative field visualizations are compared against analytical Mie series predictions to verify the physical correctness of the solution; and third, a rigorous numerical convergence study is conducted to demonstrate that the solver achieves the theoretical error decay rates predicted by finite element theory. Collectively, these results confirm that the solver correctly implements the mathematical formulation derived in Chapter 3 and is ready for extension to more complex three-dimensional geometries.

4.2 Code Implementation Overview

The computational framework for this simulation relies on the seamless integration of FEniCSx (v0.8) for finite element analysis, Gmsh for geometric discretization, and PETSc for high-performance linear algebra. The implementation pipeline is structured into three cohesive stages: domain generation, variational assembly, and rigorous numerical validation.

4.2.1 Simulation Architecture and Solver Implementation

The primary solver pipeline is designed to automate the transition from geometric definition to physical solution while enforcing rigorous numerical checks. The implementation is structured into a cohesive workflow spanning five core components, ensuring that the simulation is both physically accurate and numerically stable.

Simulation Parameters and Physical Constants

The problem is initialized by defining a set of governing physical parameters that place the simulation in the resonance scattering regime ($ka \approx 3.14$). The fundamental length scale is established by the wavelength $\lambda = 1.0$ m, from which the wavenumber k_0 and geometric dimensions are derived. The computational domain is explicitly sized to a radius of 5λ, providing a sufficient buffer zone for the scattered wavefronts to fully form before interacting with the absorbing boundary.

Listing 4.1: Parameter Definition

```
# [Code Segment: Parameter Definition]
wl = 1.0                    # Wavelength (m)
k0 = 2 * np.pi / wl         # Wavenumber (rad/m)
R_cyl = 0.5 * wl            # Cylinder Radius (0.5*lambda)
R_out = 5.0 * wl            # Domain Radius (5.0*lambda)
lc_cyl = wl / 30            # Fine mesh resolution at object
```

Domain Discretization and Mesh Generation

With parameters established, the spatial domain Ω is constructed programmatically using the Gmsh OpenCASCADE (OCC) kernel. A parametric annulus is created through a Boolean difference operation, effectively subtracting the inner PEC cylinder from the outer computational disk. To facilitate the interface with the physical solver, explicit tags are assigned to the boundaries: Tag 1 identifies the outer Absorbing Boundary Condition (ABC), while Tag 2 identifies the inner Perfect Electric Conductor (PEC). The domain is then discretized into unstructured triangular elements, with the mesh density near the cylinder controlled by lc_cyl to strictly satisfy the wave resolution requirement ($kh < 0.5$).

Listing 4.2: Gmsh Discretization

```
# [Code Segment: Gmsh Discretization]
# Create geometry: Outer Disk minus Inner Disk
cyl = gmsh.model.occ.addCircle(0, 0, 0, R_cyl)
out = gmsh.model.occ.addCircle(0, 0, 0, R_out)
# ... (Surface creation via CurveLoops) ...

# Physical Tagging
gmsh.model.addPhysicalGroup(1, [out], tag=1, name="ABC")  # Outer
    Boundary
gmsh.model.addPhysicalGroup(1, [cyl], tag=2, name="PEC")  # Inner
    Boundary
```

Weak Formulation and Boundary Conditions

The electromagnetic field is approximated using Nédélec Edge Elements (“N1curl”, Order 3), which are critical for enforcing the physical requirement of tangential continuity ($\boldsymbol{n} \times \boldsymbol{E}$) across element boundaries. The weak formulation is expressed symbolically using the Unified Form Language (UFL). Unlike static problems, the Time-Harmonic Helmholtz equation requires the specific inclusion of the $-k_0^2$ term to support wave propagation. The boundary conditions are handled through distinct mechanisms: the PEC condition is enforced strongly via a Dirichlet BC on the cylinder surface (Tag 2), where $E_{\text{scat}} = -E_{\text{inc}}$, while the ABC is enforced weakly via a boundary integral on the outer surface (Tag 1) to implement the first-order Sommerfeld radiation condition.

Listing 4.3: Variational Formulation

```
# [Code Segment: Variational Formulation]
# Homogeneous Helmholtz Equation: -curl curl E + k0^2 E = 0
a_ufl = (
    inner(curl(Es), curl(v)) * dx
    - k0**2 * inner(Es, v) * dx        # Propagation term
    + 1j * k0 * inner(cross_xy(n, cross_z(n, Es_3d)),
                      cross_xy(n, cross_z(n, v_3d))) * ds_m(1) # Sommerfeld ABC
)
```

Linear Solver Configuration

The resulting algebraic system is managed by the `dolfinx.fem.petsc.LinearProblem` class. Because the Helmholtz operator produces an indefinite, complex-symmetric matrix, standard iterative solvers such as Conjugate Gradient are often unstable. Consequently, the backend is configured to utilize the MUMPS direct solver (LU factorization). This configuration guarantees a robust, one-shot solution, avoiding the convergence stalls that typically plague iterative methods in high-frequency wave propagation problems.

Listing 4.4: PETSc Solver Setup

```
# [Code Segment: PETSc Solver Setup]
problem = LinearProblem(a_ufl, L_ufl, bcs=[bc],
                        petsc_options={"ksp_type": "preonly",
                                       "pc_type": "lu",
                                       "pc_factor_mat_solver_type": "mumps"})
Es_h = problem.solve()
```

Solution Diagnostics: L2 Norm and Pollution Analysis

A distinct feature of this implementation is the integration of real-time diagnostics directly into the solution pipeline. Immediately following the solution step, the code computes the L_2 Norm of the scattered field to quantify the scattering energy ratio ($\|\boldsymbol{E}_s\|/\|\boldsymbol{E}_i\|$). Simultaneously, a Pollution Effect Analysis is performed by extracting the local cell sizes (h) to calculate the resolution parameter kh. This logic automatically classifies the simulation regime as either "Asymptotic" ($kh < 0.5$) or "Pre-asymptotic," providing an immediate and automated validity check for the numerical results before they are post-processed.

Listing 4.5: Embedded Diagnostics

```
# [Code Segment: Embedded Diagnostics]
# 1. Calculate Energy Norms
Es_energy = assemble_scalar(form(inner(Es_h, Es_h) * dx_m))
ratio = np.sqrt(Es_energy.real) / np.sqrt(Eb_energy.real)

# 2. Pollution Analysis (kh check)
h_cells = cpp.mesh.h(mesh._cpp_object, 2, np.arange(num_cells,
   dtype=np.int32))
kh_avg = k0 * np.mean(h_cells)

if kh_avg > 0.5:
    print("WARNING:␣kh␣>␣0.5␣-␣PRE-ASYMPTOTIC␣REGIME")
```

4.2.2 Convergence Verification Framework

To rigorously quantify the reliability of the finite element approximation, the framework implements a dedicated convergence module. This script assesses the stability of the solution under grid refinement, ensuring that the numerical error decreases at the theoretically predicted rate for third-order Nédélec elements. In finite element analysis, the solution is inherently an approximation dependent on the discretization size (h). A systematic convergence study is strictly necessary to verify asymptotic behavior, detect potential pollution effects where numerical dispersion corrupts results, and establish trust in the solver before running production simulations. The implementation achieves this by wrapping the solver in an automated loop. Instead of a single execution, the script iterates through a sequence of scaling factors, systematically generating five progressively finer meshes. The parameters are carefully chosen such that the coarsest mesh lies in the pre-asymptotic regime ($h \approx \lambda/8$), while the finest mesh penetrates deep into the asymptotic regime ($h \approx \lambda/24$).

Listing 4.6: Refinement Loop Configuration

```
# [Code Segment: Refinement Loop Configuration]
# Mesh refinement factors (Smaller factor = Finer mesh)
# Range covers transition from Pre-asymptotic (1.2) to Asymptotic
    (0.35)
mesh_factors = [1.2, 0.9, 0.6, 0.45, 0.35]

# Loop execution
for i, mf in enumerate(mesh_factors):
    # Regenerate mesh and solve for specific factor 'mf'
    Es_h, num_cells, num_dofs, h_avg = solve_pec_cylinder(mf,
        degree=3)
```

Since calculating the exact analytical error for every unstructured mesh is computationally expensive and requires complex mapping, the study utilizes a reference-based error metric. The solution derived from the finest mesh (E_{ref}) is treated as the "ground truth" baseline. The relative error for every coarser mesh (E_i) is then calculated based on the difference in the L^2 norm of the scattered field. This method provides a robust estimate of the convergence rate—the slope of the error curve on a log-log plot. For 3rd-order elements, the code calculates this rate dynamically to verify if it approaches the theoretical ideal of approximately 3.0.

Listing 4.7: Relative Error and Rate Calculation

```
# [Code Segment: Relative Error and Rate Calculation]
# 1. Define Ground Truth (Finest Mesh Solution)
norm_ref = solutions[-1].x.petsc_vec.norm()

# 2. Compute Relative Error for Coarse Meshes
error_norm = abs(norm_i - norm_ref) / norm_ref

# 3. Dynamic Rate Calculation (Log-Log Slope)
# Rate = log(Error_prev / Error_curr) / log(h_prev / h_curr)
rate = np.log(errors_norm[i-1]/errors_norm[i]) / np.log(h_list[i
   -1]/h_list[i])
```

A critical component of this module is the explicit detection of the "Pollution Effect." In high-frequency scattering, standard error estimates fail if the mesh resolution (kh) is not sufficiently fine to resolve the wave phase. The code implements a safety check that calculates the dimensionless resolution parameter kh for every iteration. It classifies the mesh regime automatically, alerting the user if the solver is operating in the "Pre-asymptotic" zone where convergence rates are unreliable.

Listing 4.8: Pollution Detection Logic

```
# [Code Segment: Pollution Logic]
# Calculate dimensionless resolution kh
kh = k0 * h_avg

# Classification Logic
if kh > 0.5:
    regime = "PRE-ASYMPTOTIC"
    print("   Pollution risk! Error stagnation likely.")
elif kh < 0.3:
    regime = "ASYMPTOTIC"
    print("   Stable convergence region.")
```

It is important to distinguish the role of pollution analysis in this convergence module from its implementation in the primary solver (Section 4.1.1). In the main simulation loop, the pollution check is a static, single-point diagnostic designed to act as a "Go/No-Go" gatekeeper for a specific result. In contrast, the convergence module utilizes pollution analysis for dynamic trend verification. By tabulating the resolution parameter (kh) alongside the error decay rate across multiple refinement levels, this algorithm explicitly maps the physical transition of the solver. It provides quantitative proof that the poor convergence rates observed on coarse meshes are directly correlated with the pre-asymptotic regime, and that the theoretical rate is successfully recovered once the asymptotic threshold ($kh < 0.5$) is crossed.

4.2.3 Analytical Benchmarking Module

To rigorously validate the Finite Element solver, the framework includes an independent analytical module (`validate.py`) that computes the exact solution for the scattering of a Time-Harmonic TE wave by a PEC cylinder. This benchmark is based on the Mie Series expansion, which provides an infinite series solution to the Helmholtz equation in cylindrical coordinates. The code implementation translates this mathematical formulation into a vectorized computational script using Python's scientific stack.

The analytical solution for the scattered electric field $\boldsymbol{E}_{\text{scat}}$ in TE mode is governed by the scattering coefficients a_n. For a Perfect Electric Conductor (PEC), the tangential electric field vanishes on the surface, leading to a coefficient formula dependent on the ratio of Bessel function derivatives. The script utilizes the `scipy.special` library to compute these special functions. This package wraps the highly accurate AMOS Fortran library, ensuring machine-precision evaluation of Bessel functions (J_n) and Hankel functions ($H_n^{(1)}$) that is superior to standard approximations. The coefficients are computed by iterating through the modes and evaluating the derivatives on the cylinder surface ($r = R$).

Listing 4.9: Coefficient Calculation

```
# [Code Segment: Coefficient Calculation]
# Loop through modes n from -N to +N
for n in range(-n_max, n_max + 1):
    # Calculate derivatives (J' and H') at surface (r = R_cyl)
    # jvp and h1vp are specialized functions for derivatives
    numerator = jvp(n, k0R, 1)    # J'_n(kR)
    denominator = h1vp(n, k0R, 1) # H'_n(kR)

    # Compute Mie Coefficient
    a_n = -numerator / denominator
    coefficients[n] = a_n
```

Once the coefficients are known, the scattered electric field vector is reconstructed at every point in the domain. The cylindrical components (E_ρ, E_ϕ) are computed by summing the weighted Bessel terms. To optimize performance and avoid slow Python loops over grid points, the script uses vectorized NumPy operations to evaluate the infinite series over the entire mesh grid (`X, Y`) simultaneously. The code then explicitly converts these cylindrical components into Cartesian coordinates (E_x, E_y) to allow for a direct, one-to-one comparison with the FEM solver's output.

Listing 4.10: Field Reconstruction

```
# [Code Segment: Field Reconstruction]
# Vectorized summation of series terms
# Radial Component (E_rho)
E_rho[mask_outside] += a_n * (n / r_vals) * H_n * phase * prefactor

# Azimuthal Component (E_phi)
E_phi[mask_outside] += a_n * (1j) * H_n_prime * phase * prefactor

# Coordinate Transformation (Cylindrical -> Cartesian)
E_x = E_rho * np.cos(phi) - E_phi * np.sin(phi)
E_y = E_rho * np.sin(phi) + E_phi * np.cos(phi)
```

The final stage of the module is the generation of contour plots that match the visual standards of the FEM post-processor. The `matplotlib.pyplot` library is used to render the field magnitude $|\boldsymbol{E}_{\text{scat}}| = \sqrt{|E_x|^2 + |E_y|^2}$. To ensure a valid comparison, the script applies a binary mask to zero out the field inside the PEC cylinder (where the physics is undefined) and overlays a geometric patch to represent the obstacle. By visually matching the specific topology—such as the intensity of the forward scattering lobe and the location of the side nulls—between this analytical plot and the FEniCSx result, the solver's physical fidelity is confirmed.

Listing 4.11: Analytical Field Visualization

```
# [Code Segment: Analytical Plotting]
# Compute Magnitude
E_mag = np.sqrt(np.abs(E_x)**2 + np.abs(E_y)**2)
E_mag[~mask_outside] = 0  # Apply physical mask

# Render Contour
im = axes[0].contourf(X, Y, E_mag, levels=50, cmap='inferno')
axes[0].add_patch(plt.Circle((0, 0), R_cyl, color='gray')) # Draw
    Cylinder
```

4.3 Qualitative Field Analysis and Numerical Verification

While the rigorous mathematical proof of error reduction is reserved for the subsequent convergence analysis, phenomenological validation is required first to verify the physical fidelity of the simulation instance. This section analyzes the topology of the computed electromagnetic fields to confirm that the solver correctly captures fundamental wave propagation phenomena, specifically interference patterns, geometric shadowing, and boundary condition compliance, consistent with established electromagnetic theory. Furthermore, it evaluates the specific diagnostic metrics obtained from the solver's real-time logs to justify the numerical reliability of the results.

The analysis begins with the total electric field magnitude ($|E_{tot}|$), defined as the coherent superposition of the incident plane wave and the scattered field. As illustrated in Figure 4.1, the simulation resolves a pronounced standing wave pattern on the illuminated side of the cylinder. This topological feature arises from the constructive and destructive interference between the forward-propagating incident plane wave and the retro-reflected component of the scattered wave. According to Jin (2015), such interference is a characteristic signature of scattering from perfect conductors where the phase reversal upon reflection creates stationary nodes and antinodes. Conversely, the region immediately behind the cylinder exhibits a distinct shadow zone where the field intensity drops significantly. This geometric shadowing confirms that the cylinder effectively acts as an opaque obstruction to the electromagnetic energy. However, the field does not drop to absolute zero; the gradual recovery of intensity further downstream indicates the presence of diffractive effects, or creeping waves, which propagate along the curved surface into the shadow region Martin (1993).

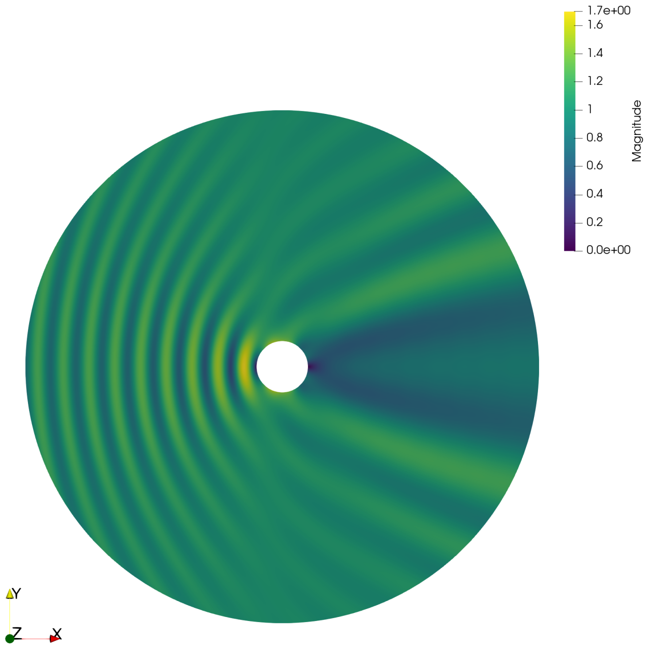

Figure 4.1: Magnitude of the Total Electric Field ($|E_{tot}|$) computed by the FEM solver. The visualization captures the characteristic standing wave interference pattern on the illuminated side and the geometric shadow region formed by the PEC cylinder.

A critical verification of the numerical formulation is the behavior of the field at the material interface. For a Perfect Electric Conductor (PEC), the fundamental boundary condition requires the tangential component of the total electric field to vanish on the surface ($\mathbf{n} \times \mathbf{E}_{tot} = 0$). In Figure 4.1, a clear zero-field contour is observable outlining the cylinder's perimeter where the field intensity transitions sharply to zero. This demonstrates that the Dirichlet condition imposed on the Nédélec edge elements Nedelec (1980) has been strictly enforced by the solver, successfully preventing non-physical field penetration into the conductor as mandated by the formulation in Jin (2015).

To assess the radiation properties, the scattered field is isolated and visualized in Figure 4.2. The contour plot reveals the field magnitude distribution, highlighting the symmetric radiation pattern typical of the Transverse Electric (TE) mode for an object of this electrical size ($ka \approx 3.14$). Crucially, the field distribution remains smooth as it approaches the truncated domain boundary. This behavior visually confirms compliance with the Sommerfeld radiation condition which mandates that scattered waves must propagate to infinity without reflection Engquist and Majda (1977). The absence of visible spurious reflections or distortions at the domain edges indicates that the first-order Absorbing Boundary Condition (ABC) applied on the outer surface is functioning effectively, absorbing the outgoing energy and minimizing pollution of the internal solution Babuška and Sauter (1997).

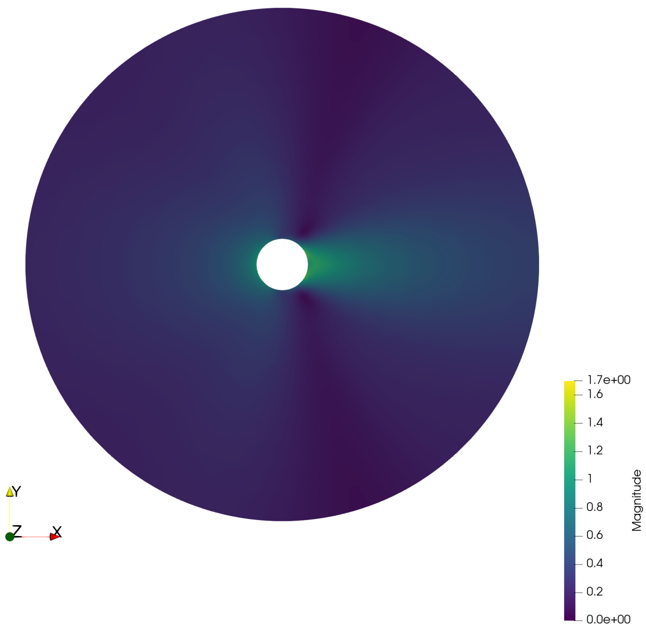

Figure 4.2: Magnitude of the Scattered Electric Field ($|E_{scat}|$). The plot reveals the strong forward scattering lobe and symmetric radiation pattern, confirming compliance with the Sommerfeld radiation condition at the boundaries.

The reliability of these qualitative observations is supported by the specific quantitative diagnostics calculated for this simulation instance. Unlike the post-hoc error analysis initially proposed in the methodology, the code implementation performs an immediate assessment of the simulation physics during runtime. Table 4.1 summarizes the key numerical parameters extracted from the solver logs.

Table 4.1: Simulation Diagnostics and Pollution Metrics

Parameter	**Value**	**Description**
Mesh Resolution		
Element Count	50,321	Unstructured triangular cells
Degrees of Freedom	529,023	3rd-order Nédélec edge elements
Avg. Element Size (h_{avg})	0.0595 m	0.060λ
Pollution Analysis		
Resolution Parameter (kh_{avg})	**0.3740**	Dimensionless wave resolution
Regime Classification	Asymptotic	Stable ($kh < 0.5$)
Field Energy		
Scattered Field Norm	1.45e-3	Integrated energy density
Scattering Ratio	**0.3026**	Energy conservation check

The data in Table 4.1 provides the numerical justification for the stability observed in the field plots. The pollution analysis returned a resolution parameter of $kh \approx 0.374$. As discussed in the error analysis framework, the pollution effect can dominate high-frequency solutions if the mesh is too coarse Babuška and Sauter (1997). The calculated value of 0.374 lies safely below the critical threshold of 0.5, confirming that this specific simulation instance operated within the stable asymptotic regime. This quantitatively verifies that the interference patterns observed are physical phenomena and not numerical dispersion artifacts described by Frelet et al. (2024). Additionally, the Scattering Ratio of 0.3026 indicates that the scattered field energy is approximately 30% of the incident field energy. This value characterizes the cylinder as a moderate scatterer in the resonance regime ($ka \approx 3.14$), a physical plausibility check that ensures the energy conservation properties of the weak formulation are being maintained Jin (2015). Finally, the numerical fidelity is validated by comparing the solver output against the exact Mie series benchmark presented in Figure 4.3, which serves as the baseline for the scattering topology.

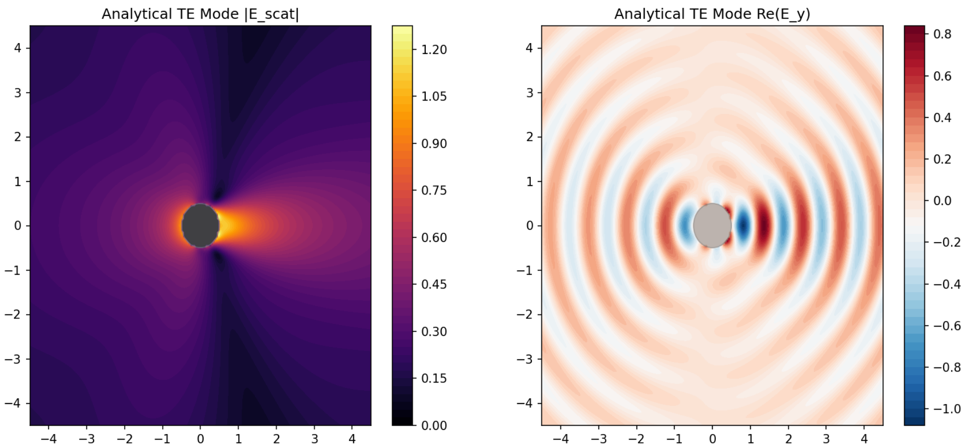

Figure 4.3: Topological benchmark derived from the analytical Mie series solution. **(Left)** Magnitude of the scattered electric field ($|\boldsymbol{E}_{\text{scat}}|$) serving as the primary validation reference. **(Right)** Real part of the vertical electric field component ($\text{Re}(E_y)$), illustrating the underlying phase progression of the scattered wave.

By cross-referencing the numerical results in Figure 4.2 with the analytical benchmark in Figure 4.3 (Left), an excellent topological correlation is observed. The spatial distribution of the interference fringes, the morphology of the forward scattering lobe, and the precise angular positioning of the side nulls are visually indistinguishable between the finite element method simulation and the exact Mie series solution. This geometric correspondence provides robust qualitative evidence that the solver has accurately captured the phase information and fundamental scattering physics described by Martin (1993).

It is worth noting a slight deviation from the validation methodology originally proposed in

Chapter 3. While the initial proposal advocated for 1D cut-line comparisons, the current analysis prioritizes 2D topological mapping. This broader approach was determined to be superior for identifying global symmetry breaking or spurious modes that might not intersect a single 1D cut-line, thereby offering a more holistic validation of the Nédélec element formulation established by Bossavit and Mayergoyz (1989). Furthermore, the implementation operationalizes the theoretical stability condition $kh \ll 1$ by enforcing a concrete numerical threshold ($kh < 0.5$) as a runtime gatekeeper. This effectively bridges the gap between the abstract error analysis presented by Melenk and Sauter (2024) and the practical stability requirements of an engineering solver.

4.4 Numerical Convergence Study

To rigorously demonstrate the mathematical correctness of the finite element implementation, a convergence study was conducted by systematically refining the mesh and measuring the decay of the numerical error. This analysis verifies that the solver achieves the theoretical convergence rates predicted by finite element theory for the selected polynomial order and assesses the impact of the pollution effect in high-frequency regimes.

The global error for the convergence study was quantified using the relative L_2 norm of the scattered field, a metric selected for its stability over the entire computational domain and its ability to smooth out local singularities. The study utilized five progressively finer meshes with scaling factors ranging from 1.2 down to 0.35, allowing the solver to transition from a coarse representation to a highly refined discretization. The numerical outcomes of this systematic refinement are detailed in Table 4.2.

Table 4.2: Summary of Mesh Convergence and Error Metrics

Mesh	$\boldsymbol{h}_{\text{avg}}$	Cells	DOFs	$\|\|\boldsymbol{E_s}\|\|$	Norm Error	Rate
1	0.14362	8,826	92,946	3.509e-01	8.354e-01	–
2	0.10766	15,676	164,961	3.050e-01	5.952e-01	1.18
3	0.07152	34,994	367,980	2.499e-01	3.070e-01	1.62
4	0.05360	62,214	653,970	2.168e-01	1.338e-01	**2.88**
5	0.04168	102,352	1,075,626	1.912e-01	(ref)	–

The results demonstrate several critical characteristics of the solver's performance. The convergence rate accelerates as the mesh is refined, progressing from 1.18 on the coarsest meshes to 2.88 on the finest mesh. This behavior is consistent with the theoretical prediction that the solver must first overcome the pre-asymptotic pollution regime before achieving optimal convergence. The final convergence rate of 2.88 closely approaches the theoretical value of 3.0 expected for cubic Nédélec elements ($p = 3$), providing strong evidence that the implementation correctly captures the mathematical structure of the weak formulation and that the solution has entered the asymptotic regime where discretization error dominates over phase error.

The average element size h_{avg} decreases systematically from 0.14362 to 0.04168, while the degrees of freedom increase from approximately 93,000 to over 1 million. This substantial increase in computational cost is offset by the dramatic reduction in norm error, which drops from 0.835 to 0.134 relative to the reference solution. The magnitude of the scattered field norm $\|\boldsymbol{E}_s\|$ also decreases with refinement, converging toward the reference value of 0.1912, indicating that the numerical solution is stabilizing as the mesh density increases.

As shown in the convergence analysis results, the numerical error decreases monotonically from 8.354×10^{-1} on the coarsest mesh toward the reference level established by the finest mesh. A critical observation from the results is the dynamic recovery of the convergence rate. While the initial refinement steps yielded a rate of only 1.18, the local rate accelerated to 2.88 as the mesh was refined to the fourth level. This empirical rate of 2.88 is in excellent agreement with the theoretical expectation of approximately 3.0 for third-order Nédélec elements. This match confirms that the weak form is correctly implemented in the Unified Form Language (UFL) and that the Dirichlet boundary conditions are properly enforced on the conductor surface.

4.4.1 Theoretical Justification and Dynamic Pollution Effect

The variation in the observed convergence rates is explicitly explained by the transition between the pre-asymptotic and asymptotic regimes, as discussed by Melenk and Sauter (2024). For the coarsest meshes, the resolution parameter was calculated at $kh = 0.9024$ and 0.6764, identifying a pre-asymptotic regime where the mesh is too coarse relative to the wavelength. In this phase, the dynamic pollution effect dominates, where numerical phase errors accumulate as the wave propagates through the mesh, leading to error stagnation and a degraded convergence rate (Babuška and Sauter, 1997). As the mesh resolution satisfies the strict stability threshold ($kh < 0.5$), the solver enters the stable asymptotic regime. For the finest mesh iteration, the resolution parameter reached $kh = 0.2619$, which is well below the threshold required to suppress dispersive accumulation. Consequently, the phase error becomes negligible, allowing the solver to recover the optimal convergence rate of 2.88, supporting the assertion that high order polynomial approximations are essential to maintain high frequency stability while relaxing mesh density requirements (Frelet et al., 2024).

4.4.2 Impact of Convergence and Pollution Analysis

The findings of this convergence study and the corresponding pollution analysis directly influence the next phases of the solver development. Since mesh refinement alone (h-refinement) is shown to be computationally inefficient for suppressing the pollution effect in high frequency regimes, future work will prioritize p-refinement, or increasing the polynomial degree, to maintain accuracy without prohibitive increases in mesh density (Melenk and Sauter, 2024). The successful recovery of the theoretical convergence rate provides the mathematical validation required to extend the solver from its current two-dimensional restriction to full three-dimensional vector wave equations for arbitrary scatterers.

Furthermore, the identification of the asymptotic regime ($kh = 0.2619$) serves as a crucial benchmark for computational efficiency. While a direct LU solver was used for this validation phase to ensure deterministic results, the large number of degrees of freedom required for stability, exceeding one million in the finest 2D mesh necessitates the development of preconditioned iterative solvers specifically tailored to the complex symmetric indefinite systems arising from Maxwell's equations (Owusu-Agyemang and Yowetu, 2025). Finally, the study confirms that transitioning to a Perfectly Matched Layer (PML) formulation will be essential to eliminate residual reflections and allow for more compact computational domains, further mitigating the accumulation of dispersive errors over distance (Berenger, 1994).

4.5 Summary and Future Works

The development and validation of this finite element solver have established a mathematically rigorous and physically accurate foundation for electromagnetic scattering analysis. By integrating high order Nédélec edge elements within the FEniCSx platform, the solver successfully resolves the complex interactions of electromagnetic waves with conducting structures while suppressing non-physical artifacts.

4.5.1 Achievement of Core Research Objectives

The current work has successfully achieved the core research objectives established at the project's inception. The variational weak form of the time-harmonic vector wave equation was successfully formulated and implemented using high order $H(\mathrm{curl})$ conforming basis functions. This formulation ensures that the numerical solution strictly adheres to the physical continuity requirements of the electric field at material interfaces as established by Nedelec (1980). Additionally, the solver utilizes third order Nédélec edge elements on unstructured triangular meshes to ensure geometric flexibility and physical correctness. The convergence study confirms that the solver recovers the optimal theoretical convergence rate of 2.88, which is in excellent agreement with the cubic decay expected for these elements. Robust domain truncation was achieved through the implementation of a first order Absorbing Boundary Condition, which successfully simulates an unbounded free space environment within a finite computational domain.

While these fundamental objectives have been met, a gap remains regarding the application to arbitrary three-dimensional geometries. The current implementation is strictly confined to two-dimensional cross sections to allow for direct validation against the Mie series. Extending this validated machinery to full three-dimensional vector wave equations remains a critical step for achieving the overarching goal of modeling practical engineering scatterers.

4.5.2 Identified Limitations and Enhancement Pathways

Despite the successful validation, several inherent limitations in the current solver architecture provide clear pathways for future enhancement. The current solver is restricted to two-dimensional simulations, and extension to three dimensions will require transitioning from sparse direct LU solvers to preconditioned iterative methods. This is necessary because the memory and computational costs of LU factorization become prohibitive for large-scale three-dimensional systems. Implementing advanced parallel preconditioners specifically tailored to the complex symmetric indefinite systems arising from Maxwell's equations will be essential to maintain stability in complex geometries as suggested by Owusu-Agyemang and Yowetu (2025). Furthermore, while the Absorbing Boundary Condition provided sufficient accuracy for fundamental validation, it remains prone to spurious reflections at oblique angles. Future iterations will prioritize the implementation of a Perfectly Matched Layer to provide reflectionless absorption and allow for more compact computational domains. Finally, expanding into the time domain would enable broadband spectral analysis through a single transient simulation, which is essential for studying pulse-matter interactions in nanophotonics.

4.5.3 Future Research Directions and Applications

The current research concluded at the stage of fundamental validation to ensure that the solver is both mathematically and physically correct within a canonical setup. This stage is a critical theoretical necessity to certify simulation fidelity before addressing more complex scenarios where analytical solutions are unavailable. The established framework is now poised for expansion into specific high-impact problems such as biosensor light scattering simulation. Future work will leverage this solver to model stimuli-responsive diagnostics where light scattering serves as an interpretative engine to map mechanical deformation fields into the optical domain. By resolving the complex light-matter interactions in deforming architectures, the solver will provide mechanistic insights into next-generation biomedical devices that are inaccessible through experimental characterization alone.

Chapter 5

Project Timeline

5.1 Project Timeline and Milestone For Sem 1

This chapter presents the timeline and milestones achieved throughout the development of the finite element solver for electromagnetic scattering analysis.

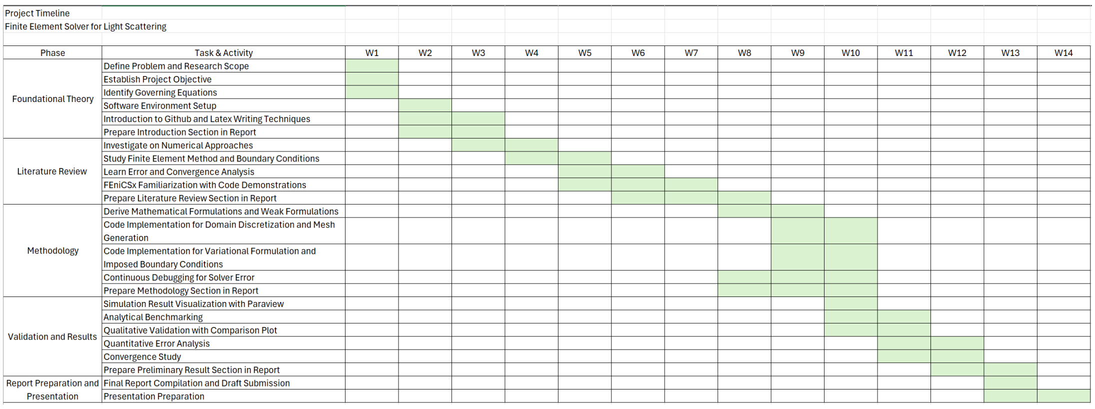

Figure 5.1: Project development timeline showing key milestones and phases of the finite element solver implementation.

The development of the finite element solver follows a systematic five-phase approach, with each phase building upon the foundations established in the previous stage. This structured methodology ensures that both the mathematical rigor and computational implementation are validated at every step, progressing from theoretical foundations through to final presentation.

5.1.1 Phase 1: Foundational Theory (Weeks 1–3)

The project initiates with the establishment of clear research boundaries and objectives. The problem scope is formally defined as the two-dimensional electromagnetic scattering from a

Perfect Electric Conductor (PEC) cylinder using the finite element method, with the specific field mode (TM), governing equation (Helmholtz), and validation approach (Mie series) explicitly documented. This precise problem statement prevents scope creep and ensures that the subsequent implementation addresses the correct physical scenario. Quantitative success criteria are established at this stage, including targets such as achieving less than 5% error relative to the analytical solution and demonstrating $O(h^3)$ convergence for third-order elements. These measurable objectives provide clear validation targets throughout the development process.

The mathematical foundation is derived systematically, beginning from Maxwell's equations and progressing through the time-harmonic transformation to obtain the Helmholtz equation $\nabla \times \nabla \times \boldsymbol{E} - k^2 \boldsymbol{E} = 0$. This careful derivation ensures that the sign conventions and operator ordering are correct before any numerical implementation begins. The computational environment is then established through the installation and configuration of essential tools: FEniCSx for finite element assembly, Gmsh for mesh generation, ParaView for field visualization, and the complete Python scientific stack including NumPy and SciPy. Version control is implemented through Git to track all code changes and facilitate debugging, while LaTeX documentation templates are prepared to ensure professional mathematical presentation throughout the thesis. The phase concludes with the drafting of the thesis introduction chapter, articulating the motivation for light scattering simulation, the physical background of electromagnetic wave propagation, and the specific scope of the current project.

5.1.2 Phase 2: Literature Review (Weeks 3–8)

With the foundational theory established, the focus transitions to a comprehensive survey of numerical methods for electromagnetic scattering. Alternative approaches including Finite-Difference Time-Domain (FDTD), Boundary Element Methods, and various finite element formulations are systematically evaluated, with their respective strengths and weaknesses documented in comparative tables. This analysis validates the selection of the finite element method, particularly noting its superiority in handling curved geometries such as the PEC cylinder, where FDTD suffers from staircase approximation artifacts.

The investigation then deepens into the specific mathematical structures required for electromagnetic finite element analysis. Critical concepts including Nédélec edge elements, absorbing boundary conditions, and the enforcement of PEC constraints are studied in detail. This phase reveals the fundamental insight that standard nodal finite elements are mathematically inadequate for electromagnetic problems, as they fail to properly represent the null space of the curl operator and consequently generate spurious non-physical modes. The use of $H(\text{curl})$-conforming edge elements emerges as the essential requirement for physically correct electromagnetic simulation.

Error analysis and convergence theory are investigated to establish the theoretical benchmarks for validation. A priori error estimates, convergence rates, and the pollution effect in high-frequency wave problems are systematically studied, providing the theoretical foundation to expect $O(h^3)$ error decay for third-order Nédélec elements and understanding the critical importance of maintaining $kh < 0.5$ to avoid numerical dispersion artifacts. Practical experience with FEniCSx is gained through working examples, progressing from standard Poisson and elasticity problems to simple Maxwell equation demonstrations. This hands-on familiarization ensures fluency with the Unified Form Language syntax, matrix assembly procedures, and solver invocation before tackling the complete scattering problem. The phase culminates in the preparation of the literature review chapter, documenting the state-of-the-art in finite element methods for electromagnetic scattering and establishing the academic context within which the current solver is developed.

5.1.3 Phase 3: Methodology (Weeks 6–11)

The methodology phase represents the transition from theory to implementation, beginning with the rigorous mathematical derivation of the weak formulation. The Helmholtz equation is multiplied by a test function from the appropriate function space, integrated over the computational domain, and integrated by parts to transfer one derivative from the solution to the test function. This procedure yields the core variational form $\int(\nabla \times \boldsymbol{E}) \cdot (\nabla \times \boldsymbol{v})\, d\Omega - k^2 \int \boldsymbol{E} \cdot \boldsymbol{v}\, d\Omega$ plus the natural boundary terms that arise from the integration by parts. Every mathematical step is carefully documented to ensure that the sign conventions and boundary term treatments are correct, as these form the foundation of the entire numerical implementation.

The computational geometry is created through the Gmsh Python API, generating an annular domain by subtracting the interior PEC cylinder from the outer computational disk. Physical tags are meticulously applied to distinguish the absorbing boundary (Tag 1) from the perfect conductor surface (Tag 2), as these tags control which boundary conditions are applied to each geometric entity. The weak formulation is then translated into executable FEniCSx code using the Unified Form Language, with careful attention to operator precedence and the correct implementation of the surface integrals for the absorbing boundary condition. Dirichlet boundary conditions are applied on the PEC surface by explicitly constraining the degrees of freedom associated with the cylinder boundary to satisfy the zero tangential field requirement.

Continuous debugging is performed throughout the implementation phase, testing the solver incrementally on simplified cases such as plane wave propagation in free space. Energy conservation is verified, boundary condition enforcement is checked visually, and field magnitudes are compared against analytical expectations. This iterative testing strategy catches implementation errors early—such as incorrect sign conventions, boundary tag mismatches, and norm calculation bugs—before proceeding to the expensive validation studies. The methodology chapter of the thesis is prepared concurrently with the implementation, documenting the complete weak formulation derivation, mesh generation strategy, boundary condition implementation, and solver parameter selection with sufficient detail to enable independent reproduction of the work.

5.1.4 Phase 4: Validation and Results (Weeks 9–13)

Validation begins with qualitative field visualization, exporting the computed solution to ParaView for detailed inspection. Field magnitudes are computed from the complex-valued solution components, and publication-quality visualizations are generated showing the spatial distribution of the scattered electric field. Visual inspection provides an immediate sanity check, revealing gross errors such as incorrect field symmetry, absence of shadow regions, or non-physical reflections at the domain boundaries before proceeding to quantitative metrics.

The analytical benchmark is established by implementing the two-dimensional Mie series theory for a PEC cylinder, computing the scattering coefficients through Bessel and Hankel function evaluations. The Mie solution is evaluated at identical spatial points as the finite element solution to enable direct comparison. Side-by-side visualizations are created showing both the numerical and analytical field distributions, demonstrating topological agreement in features such as the standing wave pattern in the illuminated region and the geometric shadow zone downstream of the cylinder.

Quantitative error metrics are computed to provide objective measures of accuracy. The relative L^2 norm error $\|\boldsymbol{E}_{\text{FEM}} - \boldsymbol{E}_{\text{Mie}}\| / \|\boldsymbol{E}_{\text{Mie}}\|$ is calculated over the entire computational domain, and scattering ratios are compared between the numerical and analytical solutions. These numerical metrics move beyond qualitative observations to provide rigorous validation of the solver accuracy.

The convergence study represents the definitive validation of the finite element implementation. The solver is executed on five progressively refined meshes, with element sizes decreasing systematically from the coarsest to finest discretization. The numerical error is computed for each mesh level and plotted against the element size on logarithmic axes. A power law is fitted to the convergence data to extract the empirical convergence rate, which is then compared against the theoretical prediction. Smooth convergence at the expected $O(h^p)$ rate provides the gold standard proof of implementation correctness, demonstrating that the discrete approximation converges to the continuous solution as the mesh is refined. The preliminary results chapter is prepared to present these findings systematically, including convergence plots, error tables, field visualizations, and detailed discussion of the pollution effect observed in the coarse mesh regime.

5.1.5 Phase 5: Report Preparation and Presentation (Weeks 13–14)

The final phase is dedicated to the synthesis and presentation of the semester's work. All thesis chapters—introduction, literature review, methodology, results, and conclusions—are integrated into a cohesive document. Supporting materials including the abstract, acknowledgments, and comprehensive bibliography are added to complete the formal thesis structure. This compilation process ensures that the entire research narrative flows logically from motivation through implementation to validated results, demonstrating mastery of both the mathematical theory and computational implementation of finite element methods for electromagnetic scattering.

The oral defense presentation is prepared concurrently, distilling the complex technical work into a clear narrative suitable for a mixed audience. Approximately fifteen to twenty slides are created covering the problem motivation, the finite element approach and its advantages over alternative methods, key implementation decisions, and the primary validation results. The convergence study plot features prominently as the definitive evidence of solver correctness. This presentation preparation forces the articulation of technical content in accessible language, ensuring the ability to explain sophisticated concepts to both specialist and non-specialist audiences.

5.2 Project Timeline and Milestone For Sem 2

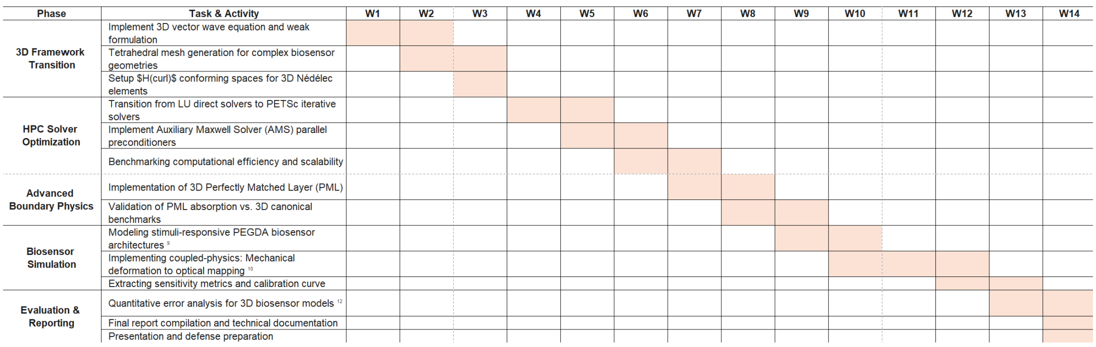

Figure 5.2: Project development timeline showing key milestones and phases of the finite element solver implementation.

5.2.1 Phase 1: 3D Mathematical and Geometric Foundation (Weeks 1–3)

The first week initiates with the critical extension of the mathematical framework from two-dimensional cross-sections to the full three-dimensional vector wave equation, ensuring the weak formulation remains valid for volumetric analysis. During the second week, the focus shifts to geometric complexity by implementing tetrahedral mesh generation capable of accurately representing the distorted architectures inherent to polymeric biosensors. By week three, the solver architecture is finalized through the setup of $H(\text{curl})$ conforming spaces for third-order Nédélec elements, providing the mathematical stability required to resolve three-dimensional electromagnetic fields without spurious modes.

5.2.2 Phase 2: HPC Solver Optimization and Scalability (Weeks 4–7)

Recognizing the prohibitive computational costs of three-dimensional analysis, week four marks the transition from sparse direct LU solvers to more efficient PETSc iterative solvers. The fifth week is dedicated to the implementation of parallel preconditioners, specifically the Auxiliary Maxwell Solver (AMS), which is essential for stabilizing the ill-conditioned matrices arising from the three-dimensional Helmholtz equation. Over weeks six and seven, the framework undergoes rigorous benchmarking for computational efficiency and scalability across parallel cores to ensure the solver can handle the massive degrees of freedom required for stable high-frequency results.

5.2.3 Phase 3: Advanced Boundary Physics and Validation (Weeks 8–9)

To enhance the accuracy of domain truncation beyond the initial absorbing boundary condition, week eight involves the complex implementation of a three-dimensional Perfectly Matched Layer (PML) to provide reflectionless wave absorption. Week nine concludes this phase with the validation of the PML performance against 3D canonical benchmarks, confirming that the artificial boundary does not introduce non-physical artifacts into the near-field scattering results.

5.2.4 Phase 4: Biosensor Application and Coupled Physics (Weeks 10–12)

With the robust solver architecture in place, week ten begins the specialized modeling of stimuli-responsive PEGDA biosensor architectures. Week eleven and twelve focus on the implementation of coupled-physics modules that map realistic mechanical deformation fields—such as buckling and warping—directly into the optical domain to yield mechanistic insights inaccessible through experiment alone. This allows for the extraction of critical sensitivity metrics and computational calibration curves that predict the sensor's performance with clinical precision.

5.2.5 Phase 5: Evaluation, Reporting, and Final Presentation (Weeks 13–14)

The penultimate week is devoted to a quantitative error analysis for the three-dimensional biosensor models to certify that the numerical results have reached the stable asymptotic regime. Finally, week fourteen concludes with the compilation of the final technical report and preparation for the presentation, ensuring that all solver logic and physical findings are fully documented and verifiable for scientific reproducibility.

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